

CHAPTER 26

Exotic Options

Derivatives such as European and American call and put options are what are termed *plain vanilla products*. They have standard well-defined properties and trade actively. Their prices or implied volatilities are quoted by exchanges or by interdealer brokers on a regular basis. One of the exciting aspects of the over-the-counter derivatives market is the number of nonstandard products that have been created by financial engineers. These products are termed *exotic options*, or simply *exotics*. Although they usually constitute a relatively small part of its portfolio, exotics are important to a derivatives dealer because they are generally much more profitable than plain vanilla products.

Exotic products are developed for a number of reasons. Sometimes they meet a genuine hedging need in the market; sometimes there are tax, accounting, legal, or regulatory reasons why corporate treasurers, fund managers, and financial institutions find exotic products attractive; sometimes the products are designed to reflect a view on potential future movements in particular market variables; occasionally an exotic product is designed by a derivatives dealer to appear more attractive than it is to an unwary corporate treasurer or fund manager.

In this chapter, we describe some of the more commonly occurring exotic options and discuss their valuation. We assume that the underlying asset provides a yield at rate q. As discussed in Chapters 17 and 18, for an option on a stock index q should be set equal to the dividend yield on the index, for an option on a currency it should be set equal to the foreign risk-free rate, and for an option on a futures contract it should be set equal to the domestic risk-free rate. Many of the options discussed in this chapter can be valued using the DerivaGem software.

26.1 PACKAGES

A *package* is a portfolio consisting of standard European calls, standard European puts, forward contracts, cash, and the underlying asset itself. We discussed a number of different types of packages in Chapter 12: bull spreads, bear spreads, butterfly spreads, calendar spreads, straddles, strangles, and so on.

Often a package is structured by traders so that it has zero cost initially. An example is a *range forward contract*.[1] This was discussed in Section 17.2. It consists of a long call and a short put or a short call and a long put. The call strike price is greater than the put strike price and the strike prices are chosen so that the value of the call equals the value of the put.

It is worth noting that any derivative can be converted into a zero-cost product by deferring payment until maturity. Consider a European call option. If c is the cost of the option when payment is made at time zero, then $A = ce^{rT}$ is the cost when payment is made at time T, the maturity of the option. The payoff is then $\max(S_T - K, 0) - A$ or $\max(S_T - K - A, -A)$. When the strike price, K, equals the forward price, other names for a deferred payment option are break forward, Boston option, forward with optional exit, and cancelable forward.

26.2 PERPETUAL AMERICAN CALL AND PUT OPTIONS

The differential equation that must be satisfied by the price of a derivative when there is a dividend at rate q is equation (17.6):

$$\frac{\partial f}{\partial t} + (r - q)S\frac{\partial f}{\partial S} + \tfrac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} = rf$$

Consider a derivative that pays off a fixed amount Q when $S = H$ for the first time. If $S < H$, the boundary conditions for the differential equation are that $f = Q$ when $S = H$ and $f = 0$ when $S = 0$. The solution $f = Q(S/H)^\alpha$ satisfies the boundary conditions when $\alpha > 0$. Furthermore, it satisfies the differential equation when

$$(r - q)\alpha + \tfrac{1}{2}\alpha(\alpha - 1)\sigma^2 = r$$

The positive solution to this equation is $\alpha = \alpha_1$, where

$$\alpha_1 = \frac{-w + \sqrt{w^2 + 2\sigma^2 r}}{\sigma^2}$$

and $w = r - q - \sigma^2/2$. It follows that the value of the derivative must be $Q(S/H)^{\alpha_1}$ because this satisfies the boundary conditions and the differential equation.

Consider next a perpetual American call option with strike price K. If the option is exercised when $S = H$, the payoff is $H - K$ and from the result just proved the value of the option is $(H - K)(S/H)^{\alpha_1}$. The holder of the call option can choose the asset price, H, at which the option is exercised. The optimal H is the one that maximizes the value we have just calculated. Using standard calculus methods, it is $H = H_1$, where

$$H_1 = K\frac{\alpha_1}{\alpha_1 - 1}$$

[1] Other names used for a range forward contract are zero-cost collar, flexible forward, cylinder option, option fence, min–max, and forward band.

The price of a perpetual call if $S < H_1$ is therefore

$$(H_1 - K)\left(\frac{S_0}{H_1}\right)^{\alpha_1} = \frac{K}{\alpha_1 - 1}\left(\frac{\alpha_1 - 1}{\alpha_1}\frac{S}{K}\right)^{\alpha_1}$$

If $S > H_1$, the call should be exercised immediately and is worth $S - K$.

To value an American put, we consider a derivative that pays off Q when $S = H$ in the situation where $S > H$ (so that the barrier H is reached from above). In this case, the boundary conditions for the differential equation are that $f = Q$ when $S = H$ and $f = 0$ as S tends to infinity. In this case, the solution $f = Q(S/H)^{-\alpha}$ satisfies the boundary conditions when $\alpha > 0$. As above, we can show that it also satisfies the differential equation when $\alpha = \alpha_2$, where

$$\alpha_2 = \frac{w + \sqrt{w^2 + 2\sigma^2 r}}{\sigma^2}$$

If the holder of the American put chooses to exercise when $S = H$, the value of the put is $(K - H)(S/H)^{-\alpha_2}$. The holder of the put will choose the exercise level $H = H_2$ to maximize this. This is

$$H_2 = K\frac{\alpha_2}{\alpha_2 + 1}$$

The price of a perpetual put if $S > H_2$ is therefore

$$(K - H_2)\left(\frac{S_0}{H_2}\right)^{-\alpha_2} = \frac{K}{\alpha_2 + 1}\left(\frac{\alpha_2 + 1}{\alpha_2}\frac{S}{K}\right)^{-\alpha_2}$$

If $S < H_2$, the put should be exercised immediately and is worth $K - S$.

Section 15.6 and Problem 15.21 give particular cases of the results here for $q = 0$.

26.3 NONSTANDARD AMERICAN OPTIONS

In a standard American option, exercise can take place at any time during the life of the option and the exercise price is always the same. The American options that are traded in the over-the-counter market sometimes have nonstandard features. For example:

1. Early exercise may be restricted to certain dates. The instrument is then known as a *Bermudan option*. (Bermuda is between Europe and America!)
2. Early exercise may be allowed during only part of the life of the option. For example, there may be an initial "lock out" period with no early exercise.
3. The strike price may change during the life of the option.

The warrants issued by corporations on their own stock often have some or all of these features. For example, in a 7-year warrant, exercise might be possible on particular dates during years 3 to 7, with the strike price being \$30 during years 3 and 4, \$32 during the next 2 years, and \$33 during the final year.

Nonstandard American options can usually be valued using a binomial tree. At each node, the test (if any) for early exercise is adjusted to reflect the terms of the option.

26.4 GAP OPTIONS

A gap call option is a European call options that pays off $S_T - K_1$ when $S_T > K_2$. The difference between a gap call option and a regular call option with a strike price of K_2 is that the payoff when $S_T > K_2$ is increased by $K_2 - K_1$. (This increase is positive or negative depending on whether $K_2 > K_1$ or $K_1 > K_2$.)

A gap call option can be valued by a small modification to the Black–Scholes–Merton formula. With our usual notation, the value is

$$S_0 e^{-qT} N(d_1) - K_1 e^{-rT} N(d_2) \tag{26.1}$$

where

$$d_1 = \frac{\ln(S_0/K_2) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

The price in this formula is greater than the price given by the Black–Scholes–Merton formula for a regular call option with strike price K_2 by

$$(K_2 - K_1)e^{-rT}N(d_2)$$

To understand this difference, note that the probability that the option will be exercised is $N(d_2)$ and, when it is exercised, the payoff to the holder of the gap option is greater than that to the holder of the regular option by $K_2 - K_1$.

For a gap put option, the payoff is $K_1 - S_T$ when $S_T < K_2$. The value of the option is

$$K_1 e^{-rT} N(-d_2) - S_0 e^{-qT} N(-d_1) \tag{26.2}$$

where d_1 and d_2 are defined as for equation (26.1).

Example 26.1

An asset is currently worth \$500,000. Over the next year, its price is expected to have a volatility of 20%. The risk-free rate is 5%, and no income is expected. Suppose that an insurance company agrees to buy the asset for \$400,000 if its value has fallen below \$400,000 at the end of one year. The payout will be $400{,}000 - S_T$ whenever the value of the asset is less than \$400,000. The insurance company has provided a regular put option where the policyholder has the right to sell the asset to the insurance company for \$400,000 in one year. This can be valued using equation (15.21), with $S_0 = 500{,}000, K = 400{,}000, r = 0.05, \sigma = 0.2, T = 1$. The value is \$3,436.

Suppose next that the cost of transferring the asset is \$50,000 and this cost is borne by the policyholder. The option is then exercised only if the value of the asset is less than \$350,000. In this case, the cost to the insurance company is $K_1 - S_T$ when $S_T < K_2$, where $K_2 = 350{,}000$, $K_1 = 400{,}000$, and S_T is the price of the asset in one year. This is a gap put option. The value is given by equation (26.2), with $S_0 = 500{,}000$, $K_1 = 400{,}000$, $K_2 = 350{,}000$, $r = 0.05$, $q = 0$, $\sigma = 0.2$, $T = 1$. It is \$1,896. Recognizing the costs to the policyholder of making a claim reduces the cost of the policy to the insurance company by about 45% in this case.

26.5 FORWARD START OPTIONS

Forward start options are options that will start at some time in the future. Sometimes employee stock options, which were discussed in Chapter 16, can be viewed as forward start options. This is because the company commits (implicitly or explicitly) to granting at-the-money options to employees in the future.

Consider a forward start at-the-money European call option that will start at time T_1 and mature at time T_2. Suppose that the asset price is S_0 at time zero and S_1 at time T_1. To value the option, we note from the European option pricing formulas in Chapters 15 and 17 that the value of an at-the-money call option on an asset is proportional to the asset price. The value of the forward start option at time T_1 is therefore cS_1/S_0, where c is the value at time zero of an at-the-money option that lasts for $T_2 - T_1$. Using risk-neutral valuation, the value of the forward start option at time zero is

$$e^{-rT_1}\hat{E}\left[c\,\frac{S_1}{S_0}\right]$$

where $\hat{E}$ denotes the expected value in a risk-neutral world. Since c and S_0 are known and $\hat{E}[S_1] = S_0 e^{(r-q)T_1}$, the value of the forward start option is ce^{-qT_1}. For a non-dividend-paying stock, $q = 0$ and the value of the forward start option is exactly the same as the value of a regular at-the-money option with the same life as the forward start option.

26.6 CLIQUET OPTIONS

A cliquet option (which is also called a ratchet or strike reset option) is a series of call or put options with rules for determining the strike price. Suppose that the reset dates are at times $t_1, t_2, \ldots, t_{n-1}$ with t_n being the end of the cliquet's life. A simple structure would be as follows. The first option has a strike price K equal to the initial asset price and lasts between times 0 and t_1; the second option provides a payoff at time t_2 with a strike price equal to the value of the asset at time t_1; the third option provides a payoff at time t_3 with a strike price equal to the value of the asset at time t_2; and so on. This is a regular option plus $n - 1$ forward start options. The latter can be valued as described in Section 26.5.

Some cliquet options are much more complicated than the one described here. For example, sometimes there are upper and lower limits on the total payoff over the whole period; sometimes cliquets terminate at the end of a period if the asset price is in a certain range. When analytic results are not available, Monte Carlo simulation is often the best approach for valuation.

26.7 COMPOUND OPTIONS

Compound options are options on options. There are four main types of compound options: a call on a call, a put on a call, a call on a put, and a put on a put. Compound options have two strike prices and two exercise dates. Consider, for example, a call on a call. On the first exercise date, T_1, the holder of the compound option is entitled to pay the first strike price, K_1, and receive a call option. The call option gives the holder the

right to buy the underlying asset for the second strike price, K_2, on the second exercise date, T_2. The compound option will be exercised on the first exercise date only if the value of the option on that date is greater than the first strike price.

When the usual geometric Brownian motion assumption is made, European-style compound options can be valued analytically in terms of integrals of the bivariate normal distribution.[2] With our usual notation, the value at time zero of a European call option on a call option is

$$S_0 e^{-qT_2} M(a_1, b_1; \sqrt{T_1/T_2}) - K_2 e^{-rT_2} M(a_2, b_2; \sqrt{T_1/T_2}) - e^{-rT_1} K_1 N(a_2)$$

where

$$a_1 = \frac{\ln(S_0/S^*) + (r - q + \sigma^2/2)T_1}{\sigma\sqrt{T_1}}, \quad a_2 = a_1 - \sigma\sqrt{T_1}$$

$$b_1 = \frac{\ln(S_0/K_2) + (r - q + \sigma^2/2)T_2}{\sigma\sqrt{T_2}}, \quad b_2 = b_1 - \sigma\sqrt{T_2}$$

The function $M(a, b:\rho)$ is the cumulative bivariate normal distribution function that the first variable will be less than a and the second will be less than b when the coefficient of correlation between the two is ρ.[3] The variable S^* is the asset price at time T_1 for which the option price at time T_1 equals K_1. If the actual asset price is above S^* at time T_1, the first option will be exercised; if it is not above S^*, the option expires worthless.

With similar notation, the value of a European put on a call is

$$K_2 e^{-rT_2} M(-a_2, b_2; -\sqrt{T_1/T_2}) - S_0 e^{-qT_2} M(-a_1, b_1; -\sqrt{T_1/T_2}) + e^{-rT_1} K_1 N(-a_2)$$

The value of a European call on a put is

$$K_2 e^{-rT_2} M(-a_2, -b_2; \sqrt{T_1/T_2}) - S_0 e^{-qT_2} M(-a_1, -b_1; \sqrt{T_1/T_2}) - e^{-rT_1} K_1 N(-a_2)$$

The value of a European put on a put is

$$S_0 e^{-qT_2} M(a_1, -b_1; -\sqrt{T_1/T_2}) - K_2 e^{-rT_2} M(a_2, -b_2; -\sqrt{T_1/T_2}) + e^{-rT_1} K_1 N(a_2)$$

26.8 CHOOSER OPTIONS

A *chooser* option (sometimes referred to as an *as you like it* option) has the feature that, after a specified period of time, the holder can choose whether the option is a call or a put. Suppose that the time when the choice is made is T_1. The value of the chooser option at this time is

$$\max(c, p)$$

where c is the value of the call underlying the option and p is the value of the put underlying the option.

If the options underlying the chooser option are both European and have the same strike price, put–call parity can be used to provide a valuation formula. Suppose that S_1

[2] See R. Geske, "The Valuation of Compound Options," *Journal of Financial Economics*, 7 (1979): 63–81; M. Rubinstein, "Double Trouble," *Risk*, December 1991/January 1992: 53–56.

[3] See Technical Note 5 at www-2.rotman.utoronto.ca/~hull/TechnicalNotes for a numerical procedure for calculating M. A function for calculating M is also on the website.

is the asset price at time T_1, K is the strike price, T_2 is the maturity of the options, and r is the risk-free interest rate. Put–call parity implies that

$$\begin{aligned}\max(c,\ p) &= \max(c,\ c + Ke^{-r(T_2-T_1)} - S_1e^{-q(T_2-T_1)})\\ &= c + e^{-q(T_2-T_1)}\max(0, Ke^{-(r-q)(T_2-T_1)} - S_1)\end{aligned}$$

This shows that the chooser option is a package consisting of:

1. A call option with strike price K and maturity T_2
2. $e^{-q(T_2-T_1)}$ put options with strike price $Ke^{-(r-q)(T_2-T_1)}$ and maturity T_1

As such, it can readily be valued.

More complex chooser options can be defined where the call and the put do not have the same strike price and time to maturity. They are then not packages and have features that are somewhat similar to compound options.

26.9 BARRIER OPTIONS

Barrier options are options where the payoff depends on whether the underlying asset's price reaches a certain level during a certain period of time.

A number of different types of barrier options regularly trade in the over-the-counter market. They are attractive to some market participants because they are less expensive than the corresponding regular options. These barrier options can be classified as either *knock-out options* or *knock-in options*. A knock-out option ceases to exist when the underlying asset price reaches a certain barrier; a knock-in option comes into existence only when the underlying asset price reaches a barrier.

Equations (17.4) and (17.5) show that the values at time zero of a regular call and put option are

$$\begin{aligned}c &= S_0e^{-qT}N(d_1) - Ke^{-rT}N(d_2)\\ p &= Ke^{-rT}N(-d_2) - S_0e^{-qT}N(-d_1)\end{aligned}$$

where

$$\begin{aligned}d_1 &= \frac{\ln(S_0/K) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}}\\ d_2 &= \frac{\ln(S_0/K) + (r - q - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}\end{aligned}$$

A *down-and-out call* is one type of knock-out option. It is a regular call option that ceases to exist if the asset price reaches a certain barrier level H. The barrier level is below the initial asset price. The corresponding knock-in option is a *down-and-in call*. This is a regular call that comes into existence only if the asset price reaches the barrier level.

If H is less than or equal to the strike price, K, the value of a down-and-in call at time zero is

$$c_{\text{di}} = S_0e^{-qT}(H/S_0)^{2\lambda}N(y) - Ke^{-rT}(H/S_0)^{2\lambda-2}N(y - \sigma\sqrt{T})$$

where

$$\lambda = \frac{r - q + \sigma^2/2}{\sigma^2}$$

$$y = \frac{\ln[H^2/(S_0K)]}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}$$

Because the value of a regular call equals the value of a down-and-in call plus the value of a down-and-out call, the value of a down-and-out call is given by

$$c_{\text{do}} = c - c_{\text{di}}$$

If $H \geq K$, then

$$c_{\text{do}} = S_0N(x_1)e^{-qT} - Ke^{-rT}N(x_1 - \sigma\sqrt{T}) - S_0e^{-qT}(H/S_0)^{2\lambda}N(y_1) + Ke^{-rT}(H/S_0)^{2\lambda-2}N(y_1 - \sigma\sqrt{T})$$

and

$$c_{\text{di}} = c - c_{\text{do}}$$

where

$$x_1 = \frac{\ln(S_0/H)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}, \quad y_1 = \frac{\ln(H/S_0)}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}$$

An *up-and-out call* is a regular call option that ceases to exist if the asset price reaches a barrier level, H, that is higher than the current asset price. An *up-and-in call* is a regular call option that comes into existence only if the barrier is reached. When H is less than or equal to K, the value of the up-and-out call, c_{uo}, is zero and the value of the up-and-in call, c_{ui}, is c. When H is greater than K,

$$c_{\text{ui}} = S_0N(x_1)e^{-qT} - Ke^{-rT}N(x_1 - \sigma\sqrt{T}) - S_0e^{-qT}(H/S_0)^{2\lambda}[N(-y) - N(-y_1)] + Ke^{-rT}(H/S_0)^{2\lambda-2}[N(-y + \sigma\sqrt{T}) - N(-y_1 + \sigma\sqrt{T})]$$

and

$$c_{\text{uo}} = c - c_{\text{ui}}$$

Put barrier options are defined similarly to call barrier options. An *up-and-out put* is a put option that ceases to exist when a barrier, H, that is greater than the current asset price is reached. An *up-and-in put* is a put that comes into existence only if the barrier is reached. When the barrier, H, is greater than or equal to the strike price, K, their prices are

$$p_{\text{ui}} = -S_0e^{-qT}(H/S_0)^{2\lambda}N(-y) + Ke^{-rT}(H/S_0)^{2\lambda-2}N(-y + \sigma\sqrt{T})$$

and

$$p_{\text{uo}} = p - p_{\text{ui}}$$

When H is less than or equal to K,

$$p_{\text{uo}} = -S_0N(-x_1)e^{-qT} + Ke^{-rT}N(-x_1 + \sigma\sqrt{T}) + S_0e^{-qT}(H/S_0)^{2\lambda}N(-y_1) - Ke^{-rT}(H/S_0)^{2\lambda-2}N(-y_1 + \sigma\sqrt{T})$$

and

$$p_{\text{ui}} = p - p_{\text{uo}}$$

A *down-and-out put* is a put option that ceases to exist when a barrier less than the current asset price is reached. A *down-and-in put* is a put option that comes into

existence only when the barrier is reached. When the barrier is greater than the strike price, $p_{do} = 0$ and $p_{di} = p$. When the barrier is less than the strike price,

$$p_{di} = -S_0 N(-x_1)e^{-qT} + Ke^{-rT}N(-x_1 + \sigma\sqrt{T}) + S_0 e^{-qT}(H/S_0)^{2\lambda}[N(y) - N(y_1)] - Ke^{-rT}(H/S_0)^{2\lambda-2}[N(y - \sigma\sqrt{T}) - N(y_1 - \sigma\sqrt{T})]$$

and

$$p_{do} = p - p_{di}$$

All of these valuations make the usual assumption that the probability distribution for the asset price at a future time is lognormal. An important issue for barrier options is the frequency with which the asset price, S, is observed for purposes of determining whether the barrier has been reached. The analytic formulas given in this section assume that S is observed continuously and sometimes this is the case.[4] Often, the terms of a contract state that S is observed periodically; for example, once a day at 3 p.m. Broadie, Glasserman, and Kou provide a way of adjusting the formulas we have just given for the situation where the price of the underlying is observed discretely.[5] The barrier level H is replaced by $He^{0.5826\sigma\sqrt{T/m}}$ for an up-and-in or up-and-out option and by $He^{-0.5826\sigma\sqrt{T/m}}$ for a down-and-in or down-and-out option, where m is the number of times the asset price is observed (so that T/m is the time interval between observations).

Barrier options often have quite different properties from regular options. For example, sometimes vega is negative. Consider an up-and-out call option when the asset price is close to the barrier level. As volatility increases, the probability that the barrier will be hit increases. As a result, a volatility increase can cause the price of the barrier option to decrease in these circumstances.

One disadvantage of the barrier options we have considered so far is that a "spike" in the asset price can cause the option to be knocked in or out. An alternative structure is a *Parisian option*, where the asset price has to be above or below the barrier for a period of time for the option to be knocked in or out. For example, a down-and-out Parisian put option with a strike price equal to 90% of the initial asset price and a barrier at 75% of the initial asset price might specify that the option is knocked out if the asset price is below the barrier for 50 days. The confirmation might specify that the 50 days are a "continuous period of 50 days" or "any 50 days during the option's life." Parisian options are more difficult to value than regular barrier options.[6] Monte Carlo simulation and binomial trees can be used with the enhancements discussed in Sections 27.5 and 27.6.

26.10 BINARY OPTIONS

Binary or digital options are options with discontinuous payoffs. A simple example of a binary option is a *cash-or-nothing call*. This pays off nothing if the asset price ends up

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[4] One way to track whether a barrier has been reached from below (above) is to send a limit order to the exchange to sell (buy) the asset at the barrier price and see whether the order is filled.

[5] M. Broadie, P. Glasserman, and S. G. Kou, "A Continuity Correction for Discrete Barrier Options," *Mathematical Finance* 7, 4 (October 1997): 325–49.

[6] See, for example, M. Chesney, J. Cornwall, M. Jeanblanc-Picqué, G. Kentwell, and M. Yor, "Parisian pricing," *Risk*, 10, 1 (1997), 77–79.

below the strike price at time T and pays a fixed amount, Q, if it ends up above the strike price. In a risk-neutral world, the probability of the asset price being above the strike price at the maturity of an option is, with our usual notation, $N(d_2)$. The value of a cash-or-nothing call is therefore $Qe^{-rT}N(d_2)$. A *cash-or-nothing put* is defined analogously to a cash-or-nothing call. It pays off Q if the asset price is below the strike price and nothing if it is above the strike price. The value of a cash-or-nothing put is $Qe^{-rT}N(-d_2)$.

Another type of binary option is an *asset-or-nothing call*. This pays off nothing if the underlying asset price ends up below the strike price and pays the asset price if it ends up above the strike price. With our usual notation, the value of an asset-or-nothing call is $S_0e^{-qT}N(d_1)$. An *asset-or-nothing put* pays off nothing if the underlying asset price ends up above the strike price and the asset price if it ends up below the strike price. The value of an asset-or-nothing put is $S_0e^{-qT}N(-d_1)$.

Binary options have discontinuous payoffs. This can create problems if the underlying asset is thinly traded so that relatively small buy or sell trades move the price. Consider a cash-or-nothing call with a strike price of \$20 and a payoff of \$1 million. If the final asset price is \$19.99, there is no payoff; if it is \$20 or more, the payoff is \$1 million. If there is one day left in the life of the option and the price a little below \$20, it would be tempting for the holder of the cash-or-nothing call to place buy orders for the underlying asset to tip the price over \$20. A similar issue can arise with barrier options on thinly traded assets. If the price of the underlying asset is close to the barrier, one side is likely to be tempted to place buy or sell orders to ensure that the barrier is reached.

A regular European call option is equivalent to a long position in an asset-or-nothing call and a short position in a cash-or-nothing call where the cash payoff in the cash-or-nothing call equals the strike price. Similarly, a regular European put option is equivalent to a long position in a cash-or-nothing put and a short position in an asset-or-nothing put where the cash payoff on the cash-or-nothing put equals the strike price.

26.11 LOOKBACK OPTIONS

The payoffs from lookback options depend on the maximum or minimum asset price reached during the life of the option. The payoff from a *floating lookback call* is the amount that the final asset price exceeds the minimum asset price achieved during the life of the option. The payoff from a *floating lookback put* is the amount by which the maximum asset price achieved during the life of the option exceeds the final asset price.

Valuation formulas have been produced for floating lookbacks.[7] The value of a floating lookback call at time zero is

$$c_{\text{fl}} = S_0e^{-qT}N(a_1) - S_0e^{-qT}\frac{\sigma^2}{2(r-q)}N(-a_1) - S_{\min}e^{-rT}\left[N(a_2) - \frac{\sigma^2}{2(r-q)}e^{Y_1}N(-a_3)\right]$$

[7] See B. Goldman, H. Sosin, and M. A. Gatto, "Path-Dependent Options: Buy at the Low, Sell at the High," *Journal of Finance*, 34 (December 1979): 1111–27.; M. Garman, "Recollection in Tranquility," *Risk*, March (1989): 16–19.

where

$$a_1 = \frac{\ln(S_0/S_{\min}) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$a_2 = a_1 - \sigma\sqrt{T},$$

$$a_3 = \frac{\ln(S_0/S_{\min}) + (-r + q + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$Y_1 = -\frac{2(r - q - \sigma^2/2)\ln(S_0/S_{\min})}{\sigma^2}$$

and $S_{\min}$ is the minimum asset price achieved to date. (If the lookback has just been originated, $S_{\min} = S_0$.) See Problem 26.23 for the $r = q$ case.

The value of a floating lookback put is

$$p_{\text{fl}} = S_{\max}e^{-rT}\left[N(b_1) - \frac{\sigma^2}{2(r-q)}e^{Y_2}N(-b_3)\right] + S_0e^{-qT}\frac{\sigma^2}{2(r-q)}N(-b_2) - S_0e^{-qT}N(b_2)$$

where

$$b_1 = \frac{\ln(S_{\max}/S_0) + (-r + q + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$b_2 = b_1 - \sigma\sqrt{T}$$

$$b_3 = \frac{\ln(S_{\max}/S_0) + (r - q - \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$Y_2 = \frac{2(r - q - \sigma^2/2)\ln(S_{\max}/S_0)}{\sigma^2}$$

and $S_{\max}$ is the maximum asset price achieved to date. (If the lookback has just been originated, then $S_{\max} = S_0$.)

A floating lookback call is a way that the holder can buy the underlying asset at the lowest price achieved during the life of the option. Similarly, a floating lookback put is a way that the holder can sell the underlying asset at the highest price achieved during the life of the option.

Example 26.2

Consider a newly issued floating lookback put on a non-dividend-paying stock where the stock price is 50, the stock price volatility is 40% per annum, the risk-free rate is 10% per annum, and the time to maturity is 3 months. In this case, $S_{\max} = 50$, $S_0 = 50$, $r = 0.1$, $q = 0$, $\sigma = 0.4$, and $T = 0.25$, $b_1 = -0.025$, $b_2 = -0.225$, $b_3 = 0.025$, and $Y_2 = 0$, so that the value of the lookback put is 7.79. A newly issued floating lookback call on the same stock is worth 8.04.

In a fixed lookback option, a strike price is specified. For a *fixed lookback call option*, the payoff is the same as a regular European call option except that the final asset price is replaced by the maximum asset price achieved during the life of the option. For a *fixed lookback put option*, the payoff is the same as a regular European put option except that the the final asset price is replaced by the minimum asset price achieved during the life of the option. Define $S^*_{\max} = \max(S_{\max}, K)$, where as before $S_{\max}$ is the maximum asset price achieved to date and K is the strike price. Also, define p^*_{fl} as the

value of a floating lookback put which lasts for the same period as the fixed lookback call when the actual maximum asset price so far, $S_{\max}$, is replaced by $S^*_{\max}$. A put–call parity type of argument shows that the value of the fixed lookback call option, c_{fix} is given by[8]

$$c_{\text{fix}} = p^*_{\text{fl}} + S_0 e^{-qT} - Ke^{-rT}$$

Similarly, if $S^*_{\min} = \min(S_{\min}, K)$, then the value of a fixed lookback put option, p_{fix}, is given by

$$p_{\text{fix}} = c^*_{\text{fl}} + Ke^{-rT} - S_0 e^{-qT}$$

where c^*_{fl} is the value of a floating lookback call that lasts for the same period as the fixed lookback put when the actual minimum asset price so far, $S_{\min}$, is replaced by $S^*_{\min}$. This shows that the equations given above for floating lookbacks can be modified to price fixed lookbacks.

Lookbacks are appealing to investors, but very expensive when compared with regular options. As with barrier options, the value of a lookback option is liable to be sensitive to the frequency with which the asset price is observed for the purposes of computing the maximum or minimum. The formulas above assume that the asset price is observed continuously. Broadie, Glasserman, and Kou provide a way of adjusting the formulas we have just given for the situation where the asset price is observed discretely.[9]

26.12 SHOUT OPTIONS

A *shout option* is a European option where the holder can "shout" to the writer at one time during its life. At the end of the life of the option, the option holder receives either the usual payoff from a European option or the intrinsic value at the time of the shout, whichever is greater. Suppose the strike price is \$50 and the holder of a call shouts when the price of the underlying asset is \$60. If the final asset price is less than \$60, the holder receives a payoff of \$10. If it is greater than \$60, the holder receives the excess of the asset price over \$50.

A shout option has some of the same features as a lookback option, but is considerably less expensive. It can be valued by noting that if the holder shouts at a time τ when the asset price is S_τ the payoff from the option is

$$\max(0, S_T - S_\tau) + (S_\tau - K)$$

where, as usual, K is the strike price and S_T is the asset price at time T. The value at time τ if the holder shouts is therefore the present value of $S_\tau - K$ (received at time T) plus the value of a European option with strike price S_τ. The latter can be calculated using Black–Scholes–Merton formulas.

A shout option is valued by constructing a binomial or trinomial tree for the underlying asset in the usual way. Working back through the tree, the value of the option if the holder shouts and the value if the holder does not shout can be calculated at each node.

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[8] The argument was proposed by H. Y. Wong and Y. K. Kwok, "Sub-replication and Replenishing Premium: Efficient Pricing of Multi-state Lookbacks," *Review of Derivatives Research*, 6 (2003), 83–106.

[9] M. Broadie, P. Glasserman, and S. G. Kou, "Connecting Discrete and Continuous Path-Dependent Options," *Finance and Stochastics*, 2 (1998): 1–28.

The option's price at the node is the greater of the two. The procedure for valuing a shout option is therefore similar to the procedure for valuing a regular American option.

26.13 ASIAN OPTIONS

Asian options are options where the payoff depends on the arithmetic average of the price of the underlying asset during the life of the option. The payoff from an *average price call* is $\max(0, S_{\text{ave}} - K)$ and that from an *average price put* is $\max(0, K - S_{\text{ave}})$, where S_{ave} is the average price of the underlying asset. Average price options tend to be less expensive than regular options and are arguably more appropriate than regular options for meeting some of the needs of corporate treasurers. Suppose that a U.S. corporate treasurer expects to receive a cash flow of 100 million Australian dollars spread evenly over the next year from the company's Australian subsidiary. The treasurer is likely to be interested in an option that guarantees that the average exchange rate realized during the year is above some level. An average price put option can achieve this more effectively than regular put options.

Average price options can be valued using similar formulas to those used for regular options if it is assumed that S_{ave} is lognomal. As it happens, when the usual assumption is made for the process followed by the asset price, this is a reasonable assumption.[10] A popular approach is to fit a lognormal distribution to the first two moments of S_{ave} and use Black's model.[11] Suppose that M_1 and M_2 are the first two moments of S_{ave}. The value of average price calls and puts are given by equations (18.7) and (18.8), with

$$F_0 = M_1 \tag{26.3}$$

and

$$\sigma^2 = \frac{1}{T}\ln\left(\frac{M_2}{M_1^2}\right) \tag{26.4}$$

When the average is calculated continuously, and r, q, and σ are constant (as in DerivaGem):

$$M_1 = \frac{e^{(r-q)T} - 1}{(r-q)T} S_0$$

and

$$M_2 = \frac{2e^{[2(r-q)+\sigma^2]T} S_0^2}{(r-q+\sigma^2)(2r-2q+\sigma^2)T^2} + \frac{2S_0^2}{(r-q)T^2}\left(\frac{1}{2(r-q)+\sigma^2} - \frac{e^{(r-q)T}}{r-q+\sigma^2}\right)$$

More generally, when the average is calculated from observations at times $T_i\,(1 \le i \le m)$,

$$M_1 = \frac{1}{m}\sum_{i=1}^{m} F_i \quad \text{and} \quad M_2 = \frac{1}{m^2}\left(\sum_{i=1}^{m} F_i^2 e^{\sigma_i^2 T_i} + 2\sum_{j=1}^{m}\sum_{i=1}^{j-1} F_i F_j e^{\sigma_i^2 T_i}\right)$$

where F_i and σ_i are the forward price and implied volatility for maturity T_i. See Technical Note 27 on www-2.rotman.utoronto.ca/~hull/TechnicalNotes for a proof of this.

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[10] When the asset price follows geometric Brownian motion, the geometric average of the price is exactly lognormal and the arithmetic average is approximately lognormal.

[11] See S. M. Turnbull and L. M. Wakeman, "A Quick Algorithm for Pricing European Average Options," *Journal of Financial and Quantitative Analysis*, 26 (September 1991): 377–89.

Example 26.3

Consider a newly issued average price call option on a non-dividend-paying stock where the stock price is 50, the strike price is 50, the stock price volatility is 40% per annum, the risk-free rate is 10% per annum, and the time to maturity is 1 year. In this case, $S_0 = 50, K = 50, r = 0.1, q = 0, \sigma = 0.4$, and $T = 1$. If the average is calculated continuously, $M_1 = 52.59$ and $M_2 = 2{,}922.76$. From equations (26.3) and (26.4), $F_0 = 52.59$ and $\sigma = 23.54\%$. Equation (18.7), with $K = 50, T = 1$, and $r = 0.1$, gives the value of the option as 5.62. When 12, 52, and 250 observations are used for the average, the price is 6.00, 5.70, and 5.63, respectively.

We can modify the analysis to accommodate the situation where the option is not newly issued and some prices used to determine the average have already been observed. Suppose that the averaging period is composed of a period of length t_1 over which prices have already been observed and a future period of length t_2 (the remaining life of the option). Suppose that the average asset price during the first time period is $\bar{S}$. The payoff from an average price call is

$$\max\left(\frac{\bar{S}t_1 + S_{\text{ave}}t_2}{t_1 + t_2} - K, 0\right)$$

where S_{ave} is the average asset price during the remaining part of the averaging period. This is the same as

$$\frac{t_2}{t_1 + t_2}\max(S_{\text{ave}} - K^*, 0)$$

where

$$K^* = \frac{t_1 + t_2}{t_2}K - \frac{t_1}{t_2}\bar{S}$$

When $K^* > 0$, the option can be valued in the same way as a newly issued Asian option provided that we change the strike price from K to K^* and multiply the result by $t_2/(t_1 + t_2)$. When $K^* < 0$ the option is certain to be exercised and can be valued as a forward contract. The value is

$$\frac{t_2}{t_1 + t_2}[M_1 e^{-rt_2} - K^* e^{-rt_2}]$$

Another type of Asian option is an average strike option. An *average strike call* pays off $\max(0, S_T - S_{\text{ave}})$ and an *average strike put* pays off $\max(0, S_{\text{ave}} - S_T)$. Average strike options can guarantee that the average price paid for an asset in frequent trading over a period of time is not greater than the final price. Alternatively, it can guarantee that the average price received for an asset in frequent trading over a period of time is not less than the final price. It can be valued as an option to exchange one asset for another when S_{ave} is assumed to be lognormal.

26.14 OPTIONS TO EXCHANGE ONE ASSET FOR ANOTHER

Options to exchange one asset for another (sometimes referred to as *exchange options*) arise in various contexts. An option to buy yen with Australian dollars is, from the point of view of a U.S. investor, an option to exchange one foreign currency asset for

another foreign currency asset. A stock tender offer is an option to exchange shares in one stock for shares in another stock.

Consider a European option to give up an asset worth U_T at time T and receive in return an asset worth V_T. The payoff from the option is

$$\max(V_T - U_T, 0)$$

A formula for valuing this option was first produced by Margrabe.[12] Suppose that the asset prices, U and V, both follow geometric Brownian motion with volatilities σ_U and σ_V. Suppose further that the instantaneous correlation between U and V is ρ, and the yields provided by U and V are q_U and q_V, respectively. The value of the option at time zero is

$$V_0 e^{-q_V T} N(d_1) - U_0 e^{-q_U T} N(d_2) \tag{26.5}$$

where

$$d_1 = \frac{\ln(V_0/U_0) + (q_U - q_V + \hat{\sigma}^2/2)T}{\hat{\sigma}\sqrt{T}}, \quad d_2 = d_1 - \hat{\sigma}\sqrt{T}$$

and

$$\hat{\sigma} = \sqrt{\sigma_U^2 + \sigma_V^2 - 2\rho\sigma_U\sigma_V}$$

and U_0 and V_0 are the values of U and V at times zero.

This result will be proved in Chapter 28. It is interesting to note that equation (26.5) is independent of the risk-free rate r. This is because, as r increases, the growth rate of both asset prices in a risk-neutral world increases, but this is exactly offset by an increase in the discount rate. The variable $\hat{\sigma}$ is the volatility of V/U. Comparisons with equation (17.4) show that the option price is the same as the price of U_0 European call options on an asset worth V/U when the strike price is 1.0, the risk-free interest rate is q_U, and the dividend yield on the asset is q_V. Mark Rubinstein shows that the American version of this option can be characterized similarly for valuation purposes.[13] It can be regarded as U_0 American options to buy an asset worth V/U for 1.0 when the risk-free interest rate is q_U and the dividend yield on the asset is q_V. The option can therefore be valued as described in Chapter 21 using a binomial tree.

An option to obtain the better or worse of two assets can be regarded as a position in one of the assets combined with an option to exchange it for the other asset:

$$\min(U_T, V_T) = V_T - \max(V_T - U_T, 0)$$
$$\max(U_T, V_T) = U_T + \max(V_T - U_T, 0)$$

26.15 OPTIONS INVOLVING SEVERAL ASSETS

Options involving two or more risky assets are sometimes referred to as *rainbow options*. One example is the bond futures contract traded on the CBOT described in Chapter 6. The party with the short position is allowed to choose between a large number of different bonds when making delivery.

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[12] See W. Margrabe, "The Value of an Option to Exchange One Asset for Another," *Journal of Finance*, 33 (March 1978): 177–86.

[13] See M. Rubinstein, "One for Another," *Risk*, July/August 1991: 30–32

Probably the most popular option involving several assets is a European *basket option*. This is an option where the payoff is dependent on the value of a portfolio (or basket) of assets. The assets are usually either individual stocks or stock indices or currencies. A European basket option can be valued with Monte Carlo simulation, by assuming that the assets follow correlated geometric Brownian motion processes. A much faster approach is to calculate the first two moments of the basket at the maturity of the option in a risk-neutral world, and then assume that value of the basket is lognormally distributed at that time. The option can then be valued using Black's model with the parameters shown in equations (26.3) and (26.4). In this case,

$$M_1 = \sum_{i=1}^{n} F_i \quad \text{and} \quad M_2 = \sum_{i=1}^{n}\sum_{j=1}^{n} F_i F_j e^{\rho_{ij}\sigma_i\sigma_j T}$$

where n is the number of assets, T is the option maturity, F_i and σ_i are the forward price and volatility of the ith asset, and ρ_{ij} is the correlation between the ith and jth asset. See Technical Note 28 at www-2.rotman.utoronto.ca/~hull/TechnicalNotes.

26.16 VOLATILITY AND VARIANCE SWAPS

A volatility swap is an agreement to exchange the realized volatility of an asset between time 0 and time T for a prespecifed fixed volatility. The realized volatility is usually calculated as described in Section 15.4 but with the assumption that the mean daily return is zero. Suppose that there are n daily observations on the asset price during the period between time 0 and time T. The realized volatility is

$$\overline{\sigma} = \sqrt{\frac{252}{n-2}\sum_{i=1}^{n-1}\left[\ln\left(\frac{S_{i+1}}{S_i}\right)\right]^2}$$

where S_i is the ith observation on the asset price. (Sometimes $n-1$ might replace $n-2$ in this formula.)

The payoff from the volatility swap at time T to the payer of the fixed volatility is $L_{\text{vol}}(\overline{\sigma} - \sigma_K)$, where L_{vol} is the notional principal and σ_K is the fixed volatility. Whereas an option provides a complex exposure to the asset price and volatility, a volatility swap is simpler in that it has exposure only to volatility.

A variance swap is an agreement to exchange the realized variance rate $\overline{V}$ between time 0 and time T for a prespecified variance rate. The variance rate is the square of the volatility ($\overline{V} = \overline{\sigma}^2$). Variance swaps are easier to value than volatility swaps. This is because the variance rate between time 0 and time T can be replicated using a portfolio of put and call options. The payoff from a variance swap at time T to the payer of the fixed variance rate is $L_{\text{var}}(\overline{V} - V_K)$, where L_{var} is the notional principal and V_K is the fixed variance rate. Often the notional principal for a variance swap is expressed in terms of the corresponding notional principal for a volatility swap using $L_{\text{var}} = L_{\text{vol}}/(2\sigma_K)$.

Valuation of Variance Swap

Technical Note 22 at www-2.rotman.utoronto.ca/~hull/TechnicalNotes shows that, for any value S^* of the asset price, the expected average variance between times 0

and T is

$$\hat{E}(\overline{V}) = \frac{2}{T}\ln\frac{F_0}{S^*} - \frac{2}{T}\left[\frac{F_0}{S^*} - 1\right] + \frac{2}{T}\left[\int_{K=0}^{S^*}\frac{1}{K^2}e^{rT}p(K)\,dK + \int_{K=S^*}^{\infty}\frac{1}{K^2}e^{rT}c(K)\,dK\right] \quad \textbf{(26.6)}$$

where F_0 is the forward price of the asset for a contract maturing at time T, $c(K)$ is the price of a European call option with strike price K and time to maturity T, and $p(K)$ is the price of a European put option with strike price K and time to maturity T.

This provides a way of valuing a variance swap.[14] The value of an agreement to receive the realized variance between time 0 and time T and pay a variance rate of V_K, with both being applied to a principal of L_{var}, is

$$L_{\text{var}}[\hat{E}(\overline{V}) - V_K]e^{-rT} \quad \textbf{(26.7)}$$

Suppose that the prices of European options with strike prices K_i $(1 \le i \le n)$ are known, where $K_1 < K_2 < \cdots < K_n$. A standard approach for implementing equation (26.6) is to set S^* equal to the first strike price below F_0 and then approximate the integrals as

$$\int_{K=0}^{S^*}\frac{1}{K^2}e^{rT}p(K)dK + \int_{K=S^*}^{\infty}\frac{1}{K^2}e^{rT}c(K)dK = \sum_{i=1}^{n}\frac{\Delta K_i}{K_i^2}e^{rT}Q(K_i) \quad \textbf{(26.8)}$$

where $\Delta K_i = 0.5(K_{i+1} - K_{i-1})$ for $2 \le i \le n-1$, $\Delta K_1 = K_2 - K_1$, $\Delta K_n = K_n - K_{n-1}$. The function $Q(K_i)$ is the price of a European put option with strike price K_i if $K_i < S^*$ and the price of a European call option with strike price K_i if $K_i > S^*$. When $K_i = S^*$, the function $Q(K_i)$ is equal to the average of the prices of a European call and a European put with strike price K_i.

Example 26.4

Consider a 3-month contract to receive the realized variance rate of an index over the 3 months and pay a variance rate of 0.045 on a principal of \$100 million. The risk-free rate is 4% and the dividend yield on the index is 1%. The current level of the index is 1020. Suppose that, for strike prices of 800, 850, 900, 950, 1,000, 1,050, 1,100, 1,150, 1,200, the 3-month implied volatilities of the index are 29%, 28%, 27%, 26%, 25%, 24%, 23%, 22%, 21%, respectively. In this case, $n = 9$, $K_1 = 800, K_2 = 850, \ldots, K_9 = 1{,}200, F_0 = 1{,}020e^{(0.04-0.01)\times 0.25} = 1{,}027.68$, and $S^* = 1{,}000$. DerivaGem shows that $Q(K_1) = 2.22, Q(K_2) = 5.22, Q(K_3) = 11.05, Q(K_4) = 21.27, Q(K_5) = 51.21, Q(K_6) = 38.94, Q(K_7) = 20.69, Q(K_8) = 9.44, Q(K_9) = 3.57$. Also, $\Delta K_i = 50$ for all i. Hence,

$$\sum_{i=1}^{n}\frac{\Delta K_i}{K_i^2}e^{rT}Q(K_i) = 0.008139$$

From equations (26.6) and (26.8), it follows that

$$\hat{E}(\overline{V}) = \frac{2}{0.25}\ln\left(\frac{1027.68}{1{,}000}\right) - \frac{2}{0.25}\left(\frac{1027.68}{1{,}000} - 1\right) + \frac{2}{0.25}\times 0.008139 = 0.0621$$

[14] See also K. Demeterfi, E. Derman, M. Kamal, and J. Zou, "A Guide to Volatility and Variance Swaps," *The Journal of Derivatives*, 6, 4 (Summer 1999), 9–32. For options on variance and volatility, see P. Carr and R. Lee, "Realized Volatility and Variance: Options via Swaps," *Risk*, May 2007, 76–83.

From equation (26.7), the value of the variance swap (in millions of dollars) is $100 \times (0.0621 - 0.045)e^{-0.04\times 0.25} = 1.69$.

Valuation of a Volatility Swap

To value a volatility swap, we require $\hat{E}(\overline{\sigma})$, where $\overline{\sigma}$ is, as before, the realized volatility between time 0 and time T. We can write

$$\overline{\sigma} = \sqrt{\hat{E}(\overline{V})}\sqrt{1 + \frac{\overline{V} - \hat{E}(\overline{V})}{\hat{E}(\overline{V})}}$$

Expanding the second term on the right-hand side in a series gives the approximation:

$$\overline{\sigma} = \sqrt{\hat{E}(\overline{V})}\left\{1 + \frac{\overline{V} - \hat{E}(\overline{V})}{2\hat{E}(\overline{V})} - \tfrac{1}{8}\left[\frac{\overline{V} - \hat{E}(\overline{V})}{\hat{E}(\overline{V})}\right]^2\right\}$$

Taking expectations,

$$\hat{E}(\overline{\sigma}) = \sqrt{\hat{E}(\overline{V})}\left\{1 - \tfrac{1}{8}\left[\frac{\text{var}(\overline{V})}{\hat{E}(\overline{V})^2}\right]\right\} \tag{26.9}$$

where $\text{var}(\overline{V})$ is the variance of $\overline{V}$. The valuation of a volatility swap therefore requires an estimate of the variance of the realized variance rate during the life of the contract. The value of an agreement to receive the realized volatility between time 0 and time T and pay a volatility of σ_K, with both being applied to a principal of L_{vol}, is

$$L_{\text{vol}}[\hat{E}(\overline{\sigma}) - \sigma_K]e^{-rT}$$

Example 26.5

For the situation in Example 26.4, consider a volatility swap where the realized volatility is received and a volatility of 23% is paid on a principal of $100 million. In this case $\hat{E}(\overline{V}) = 0.0621$. Suppose that the standard deviation of the realized variance over 3 months has been estimated as 0.01. This means that $\text{var}(\overline{V}) = 0.0001$. Equation (26.9) gives

$$\hat{E}(\overline{\sigma}) = \sqrt{0.0621}\left(1 - \tfrac{1}{8} \times \frac{0.0001}{0.0621^2}\right) = 0.2484$$

The value of the swap in (millions of dollars) is

$$100 \times (0.2484 - 0.23)e^{-0.04\times 0.25} = 1.82$$

The VIX Index

In equation (26.6), the ln function can be approximated by the first two terms in a series expansion:

$$\ln\left(\frac{F_0}{S^*}\right) = \left(\frac{F_0}{S^*} - 1\right) - \tfrac{1}{2}\left(\frac{F_0}{S^*} - 1\right)^2$$

This means that the risk-neutral expected cumulative variance is calculated as

$$\hat{E}(\overline{V})T = -\left(\frac{F_0}{S^*} - 1\right)^2 + 2\sum_{i=1}^{n}\frac{\Delta K_i}{K_i^2}e^{rT}Q(K_i) \tag{26.10}$$

Since 2004 the VIX volatility index (see Section 15.11) has been based on equation (26.10). The procedure used on any given day is to calculate $\hat{E}(\overline{V})T$ for options that trade in the market and have maturities immediately above and below 30 days. The 30-day risk-neutral expected cumulative variance is calculated from these two numbers using interpolation. This is then multiplied by 365/30 and the index is set equal to the square root of the result. More details on the calculation can be found on the CBOE website.

26.17 STATIC OPTIONS REPLICATION

If the procedures described in Chapter 19 are used for hedging exotic options, some are easy to handle, but others are very difficult because of discontinuities (see Business Snapshot 26.1). For the difficult cases, a technique known as static options replication is sometimes useful.[15] This involves searching for a portfolio of actively traded options that approximately replicates the exotic option. Shorting this position provides the hedge.[16]

The basic principle underlying static options replication is as follows. If two portfolios are worth the same on a certain boundary, they are also worth the same at all interior points of the boundary. Consider as an example a 9-month up-and-out call option on a non-dividend-paying stock where the stock price is 50, the strike price is 50, the barrier is 60, the risk-free interest rate is 10% per annum, and the volatility is 30% per annum. Suppose that $f(S, t)$ is the value of the option at time t for a stock price of S. Any boundary in (S, t) space can be used for the purposes of producing the replicating portfolio. A convenient one to choose is shown in Figure 26.1. It is defined by $S = 60$ and $t = 0.75$. The values of the up-and-out option on the boundary are given by

$$f(S, 0.75) = \max(S - 50, 0) \quad \text{when } S < 60$$
$$f(60, t) = 0 \quad \text{when } 0 \leq t \leq 0.75$$

There are many ways that these boundary values can be approximately matched using regular options. The natural option to match the first boundary is a 9-month European call with a strike price of 50. The first component of the replicating portfolio is therefore one unit of this option. (We refer to this option as option A.)

One way of matching the $f(60, t)$ boundary is to proceed as follows:

1. Divide the life of the option into N steps of length Δt
2. Choose a European call option with a strike price of 60 and maturity at time $N\Delta t$ $(= 9$ months) to match the boundary at the $\{60, (N - 1)\Delta t\}$ point
3. Choose a European call option with a strike price of 60 and maturity at time $(N - 1)\Delta t$ to match the boundary at the $\{60, (N - 2)\Delta t\}$ point

and so on. Note that the options are chosen in sequence so that they have zero value on the parts of the boundary matched by earlier options.[17] The option with a strike price

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[15] See E. Derman, D. Ergener, and I. Kani, "Static Options Replication," *Journal of Derivatives* 2, 4 (Summer 1995): 78–95.

[16] Technical Note 22 at www-2.rotman.utoronto.ca/~hull/TechnicalNotes provides an example of static replication. It shows that the variance rate of an asset can be replicated by a position in the asset and out-of-the-money options on the asset. This result, which leads to equation (26.6), can be used to hedge variance swaps.

[17] This is not a requirement. If K points on the boundary are to be matched, we can choose K options and solve a set of K linear equations to determine required positions in the options.

Business Snapshot 26.1 Is Delta Hedging Easier or More Difficult for Exotics?

As described in Chapter 19, we can approach the hedging of exotic options by creating a delta neutral position and rebalancing frequently to maintain delta neutrality. When we do this we find some exotic options are easier to hedge than plain vanilla options and some are more difficult.

An example of an exotic option that is relatively easy to hedge is an average price option where the averaging period is the whole life of the option. As time passes, we observe more of the asset prices that will be used in calculating the final average. This means that our uncertainty about the payoff decreases with the passage of time. As a result, the option becomes progressively easier to hedge. In the final few days, the delta of the option always approaches zero because price movements during this time have very little impact on the payoff.

By contrast barrier options are relatively difficult to hedge. Consider a down-and-out call option on a currency when the exchange rate is 0.0005 above the barrier. If the barrier is hit, the option is worth nothing. If the barrier is not hit, the option may prove to be quite valuable. The delta of the option is discontinuous at the barrier making conventional hedging very difficult.

of 60 that matures in 9 months has zero value on the vertical boundary that is matched by option A. The option maturing at time $i\,\Delta t$ has zero value at the point $\{60, i\,\Delta t\}$ that is matched by the option maturing at time $(i + 1)\Delta t$ for $1 \leq i \leq N - 1$.

Suppose that $\Delta t = 0.25$. In addition to option A, the replicating portfolio consists of positions in European options with strike price 60 that mature in 9, 6, and 3 months. We will refer to these as options B, C, and D, respectively. Given our assumptions

Figure 26.1 Boundary points used for static options replication example.

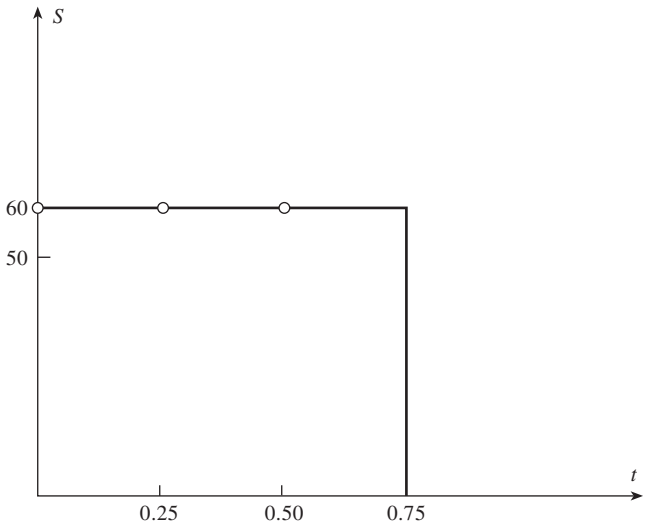


Table 26.1 The portfolio of European call options used to replicate an up-and-out option.

Option	*Strike price*	*Maturity (years)*	*Position*	*Initial value*
A	50	0.75	1.00	+6.99
B	60	0.75	−2.66	−8.21
C	60	0.50	0.97	+1.78
D	60	0.25	0.28	+0.17

about volatility and interest rates, option B is worth 4.33 at the {60, 0.5} point. Option A is worth 11.54 at this point. The position in option B necessary to match the boundary at the {60, 0.5} point is therefore $-11.54/4.33 = -2.66$. Option C is worth 4.33 at the {60, 0.25} point. The position taken in options A and B is worth −4.21 at this point. The position in option C necessary to match the boundary at the {60, 0.25} point is therefore $4.21/4.33 = 0.97$. Similar calculations show that the position in option D necessary to match the boundary at the {60, 0} point is 0.28.

The portfolio chosen is summarized in Table 26.1. (See also Sample Application F of the DerivaGem Applications.) It is worth 0.73 initially (i.e., at time zero when the stock price is 50). This compares with 0.31 given by the analytic formula for the up-and-out call earlier in this chapter. The replicating portfolio is not exactly the same as the up-and-out option because it matches the latter at only three points on the second boundary. If we use the same procedure, but match at 18 points on the second boundary (using options that mature every half month), the value of the replicating portfolio reduces to 0.38. If 100 points are matched, the value reduces further to 0.32.

To hedge a derivative, the portfolio that replicates its boundary conditions must be shorted. The hedge lasts until the end of the life of the derivative or until the boundary is reached, whichever happens first. If the boundary is reached, the hedge portfolio must be unwound and a new hedge portfolio set up.

Static options replication has the advantage over delta hedging that it does not require frequent rebalancing. It can be used for a wide range of derivatives. The user has a great deal of flexibility in choosing the boundary that is to be matched and the options that are to be used.

SUMMARY

Exotic options are options with rules governing the payoff that are more complicated than standard options. We have discussed 15 different types of exotic options: packages, perpetual American options, nonstandard American options, gap options, forward start options, cliquet options, compound options, chooser options, barrier options, binary options, lookback options, shout options, Asian options, options to exchange one asset for another, and options involving several assets. We have discussed how these can be valued using the same assumptions as those used to derive the Black–Scholes–Merton model in Chapter 15. Some can be valued analytically, but using much more complicated formulas than those for regular European calls and puts, some can be handled using analytic approximations, and some can be valued using extensions of the numerical

procedures in Chapter 21. We will present more numerical procedures for valuing exotic options in Chapter 27.

Some exotic options are easier to hedge than the corresponding regular options; others are more difficult. In general, Asian options are easier to hedge because the payoff becomes progressively more certain as we approach maturity. Barrier options can be more difficult to hedge because delta is discontinuous at the barrier. One approach to hedging an exotic option, known as static options replication, is to find a portfolio of regular options whose value matches the value of the exotic option on some boundary. The exotic option is hedged by shorting this portfolio.

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Practice Questions

26.1. Explain the difference between a forward start option and a chooser option.

26.2. Describe the payoff from a portfolio consisting of a floating lookback call and a floating lookback put with the same maturity.

26.3. Consider a chooser option where the holder has the right to choose between a European call and a European put at any time during a 2-year period. The maturity dates and strike prices for the calls and puts are the same regardless of when the choice is made. Is it ever optimal to make the choice before the end of the 2-year period? Explain your answer.

26.4. Suppose that c_1 and p_1 are the prices of a European average price call and a European average price put with strike price K and maturity T, c_2 and p_2 are the prices of a European average strike call and European average strike put with maturity T, and c_3 and p_3 are the prices of a regular European call and a regular European put with strike price K and maturity T. Show that $c_1 + c_2 - c_3 = p_1 + p_2 - p_3$.

26.5. The text derives a decomposition of a particular type of chooser option into a call maturing at time T_2 and a put maturing at time T_1. Derive an alternative decomposition into a call maturing at time T_1 and a put maturing at time T_2.

26.6. Section 26.9 gives two formulas for a down-and-out call. The first applies to the situation where the barrier, H, is less than or equal to the strike price, K. The second applies to the situation where $H \geq K$. Show that the two formulas are the same when $H = K$.

26.7. Explain why a down-and-out put is worth zero when the barrier is greater than the strike price.

26.8. Suppose that the strike price of an American call option on a non-dividend-paying stock grows at rate g. Show that if g is less than the risk-free rate, r, it is never optimal to exercise the call early.

26.9. How can the value of a forward start put option on a non-dividend-paying stock be calculated if it is agreed that the strike price will be 10% greater than the stock price at the time the option starts?

26.10. If a stock price follows geometric Brownian motion, what process does $A(t)$ follow where $A(t)$ is the arithmetic average stock price between time zero and time t?

26.11. Explain why delta hedging is easier for Asian options than for regular options.

26.12. Calculate the price of a 1-year European option to give up 100 ounces of silver in exchange for 1 ounce of gold. The current prices of gold and silver are \$1,520 and \$16, respectively; the risk-free interest rate is 10% per annum; the volatility of each commodity price is 20%; and the correlation between the two prices is 0.7. Ignore storage costs.

26.13. Is a European down-and-out option on an asset worth the same as a European down-and-out option on the asset's futures price for a futures contract maturing at the same time as the option?

26.14. Answer the following questions about compound options:

(a) What put–call parity relationship exists between the price of a European call on a call and a European put on a call? Show that the formulas given in the text satisfy the relationship.

(b) What put–call parity relationship exists between the price of a European call on a put and a European put on a put? Show that the formulas given in the text satisfy the relationship.

26.15. Does a floating lookback call become more valuable or less valuable as we increase the frequency with which we observe the asset price in calculating the minimum?

26.16. Does a down-and-out call become more valuable or less valuable as we increase the frequency with which we observe the asset price in determining whether the barrier has been crossed? What is the answer to the same question for a down-and-in call?

26.17. Explain why a regular European call option is the sum of a down-and-out European call and a down-and-in European call. Is the same true for American call options?

26.18. What is the value of a derivative that pays off \$100 in 6 months if an index is greater than 1,000 and zero otherwise? Assume that the current level of the index is 960, the risk-free rate is 8% per annum, the dividend yield on the index is 3% per annum, and the volatility of the index is 20%.

26.19. In a 3-month down-and-out call option on silver futures the strike price is \$20 per ounce and the barrier is \$18. The current futures price is \$19, the risk-free interest rate is 5%, and the volatility of silver futures is 40% per annum. Explain how the option works and calculate its value. What is the value of a regular call option on silver futures with the same terms? What is the value of a down-and-in call option on silver futures with the same terms?

26.20. A new European-style floating lookback call option on a stock index has a maturity of 9 months. The current level of the index is 400, the risk-free rate is 6% per annum, the dividend yield on the index is 4% per annum, and the volatility of the index is 20%. Use DerivaGem to value the option.

26.21. Estimate the value of a new 6-month European-style average price call option on a non-dividend-paying stock. The initial stock price is \$30, the strike price is \$30, the risk-free interest rate is 5%, and the stock price volatility is 30%.

26.22. Use DerivaGem to calculate the value of:

(a) A regular European call option on a non-dividend-paying stock where the stock price is \$50, the strike price is \$50, the risk-free rate is 5% per annum, the volatility is 30%, and the time to maturity is one year
(b) A down-and-out European call which is as in (a) with the barrier at \$45
(c) A down-and-in European call which is as in (a) with the barrier at \$45.

Show that the option in (a) is worth the sum of the values of the options in (b) and (c).

26.23. Explain adjustments that have to be made when $r = q$ for (a) the valuation formulas for floating lookback call options in Section 26.11 and (b) the formulas for M_1 and M_2 in Section 26.13.

26.24. Value the variance swap in Example 26.4 of Section 26.16 assuming that the implied volatilities for options with strike prices 800, 850, 900, 950, 1,000, 1,050, 1,100, 1,150, 1,200 are 20%, 20.5%, 21%, 21.5%, 22%, 22.5%, 23%, 23.5%, 24%, respectively.

26.25. Verify that the results in Section 26.2 for the value of a derivative that pays Q when $S = H$ are consistent with those in Section 15.6.

26.26. Consider an up-and-out barrier call option on a non-dividend-paying stock when the stock price is 50, the strike price is 50, the volatility is 30%, the risk-free rate is 5%, the time to maturity is 1 year, and the barrier at \$80. Use the DerivaGem software to value the option and graph the relationship between (a) the option price and the stock price, (b) the delta and the stock price, (c) the option price and the time to maturity, and (d) the option price and the volatility. Provide an intuitive explanation for the results you get. Show that the delta, gamma, theta, and vega for an up-and-out barrier call option can be either positive or negative.

26.27. Sample Application F in the DerivaGem Application Builder Software considers the static options replication example in Section 26.17. It shows the way a hedge can be constructed using four options (as in Section 26.17) and two ways a hedge can be constructed using 16 options.

(a) Explain the difference between the two ways a hedge can be constructed using 16 options. Explain intuitively why the second method works better.
(b) Improve on the four-option hedge by changing Tmat for the third and fourth options.
(c) Check how well the 16-option portfolios match the delta, gamma, and vega of the barrier option.

26.28. Consider a down-and-out call option on a foreign currency. The initial exchange rate is 0.90, the time to maturity is 2 years, the strike price is 1.00, the barrier is 0.80, the domestic risk-free interest rate is 5%, the foreign risk-free interest rate is 6%, and the volatility is 25% per annum. Use DerivaGem to develop a static option replication strategy involving five options.

26.29. Suppose that a stock index is currently 900. The dividend yield is 2%, the risk-free rate is 5%, and the volatility is 40%. Use the results in Technical Note 27 on the author's website to calculate the value of a 1-year average price call where the strike price is 900 and the index level is observed at the end of each quarter for the purposes of the averaging. Compare this with the price calculated by DerivaGem for a 1-year average price option where the price is observed continuously. Provide an intuitive explanation for any differences between the prices.

26.30. Use the DerivaGem Application Builder software to compare the effectiveness of daily delta hedging for (a) the option considered in Tables 19.2 and 19.3 and (b) an average price call with the same parameters. Use Sample Application C. For the average price option you will find it necessary to change the calculation of the option price in cell C16, the payoffs in cells H15 and H16, and the deltas (cells G46 to G186 and N46 to N186). Carry out 20 Monte Carlo simulation runs for each option by repeatedly pressing F9. On each run record the cost of writing and hedging the option, the volume of trading over the whole 20 weeks and the volume of trading between weeks 11 and 20. Comment on the results.

26.31. In the DerivaGem Application Builder Software modify Sample Application D to test the effectiveness of delta and gamma hedging for a call on call compound option on a 100,000 units of a foreign currency where the exchange rate is 0.67, the domestic risk-free rate is 5%, the foreign risk-free rate is 6%, the volatility is 12%. The time to maturity of the first option is 20 weeks, and the strike price of the first option is 0.015. The second option matures 40 weeks from today and has a strike price of 0.68. Explain how you modified the cells. Comment on hedge effectiveness.

26.32. Outperformance certificates (also called "sprint certificates," "accelerator certificates," or "speeders") are offered to investors by many European banks as a way of investing in a company's stock. The initial investment equals the stock price, S_0. If the stock price goes up between time 0 and time T, the investor gains k times the increase at time T, where k is a constant greater than 1.0. However, the stock price used to calculate the gain at time T is capped at some maximum level M. If the stock price goes down, the investor's loss is equal to the decrease. The investor does not receive dividends.

(a) Show that an outperformance certificate is a package.
(b) Calculate using DerivaGem the value of a one-year outperformance certificate when the stock price is 50 euros, $k = 1.5$, $M = 70$ euros, the risk-free rate is 5%, and the stock price volatility is 25%. Dividends equal to 0.5 euros are expected in 2 months, 5 months, 8 months, and 11 months.

26.33. Carry out the analysis in Example 26.4 of Section 26.16 to value the variance swap on the assumption that the life of the swap is 1 month rather than 3 months.

26.34. What is the relationship between a regular call option, a binary call option, and a gap call option?

26.35. Produce a formula for valuing a cliquet option where an amount Q is invested to produce a payoff at the end of n periods. The return earned each period is the greater of the return on an index (excluding dividends) and zero.



CHAPTER 27

More on Models and Numerical Procedures

Up to now the models we have used to value options have been based on the geometric Brownian motion model of asset price behavior that underlies the Black–Scholes–Merton formulas and the numerical procedures we have used have been relatively straightforward. In this chapter we introduce a number of new models and explain how the numerical procedures can be adapted to cope with particular situations.

Chapter 20 explained how traders overcome the weaknesses in the geometric Brownian motion model by using volatility surfaces. A volatility surface determines an appropriate volatility to substitute into Black–Scholes–Merton when pricing plain vanilla options. Unfortunately it says little about the volatility that should be used for exotic options when the pricing formulas of Chapter 26 are used. Suppose the volatility surface shows that the correct volatility to use when pricing a one-year plain vanilla option with a strike price of $40 is 27%. This is liable to be totally inappropriate for pricing a barrier option (or some other exotic option) that has a strike price of $40 and a life of one year.

The first part of this chapter discusses a number of alternatives to geometric Brownian motion that are designed to deal with the problem of pricing exotic options consistently with plain vanilla options. These alternative asset price processes fit the market prices of plain vanilla options better than geometric Brownian motion. As a result, we can have more confidence in using them to value exotic options.

The second part of the chapter extends the discussion of numerical procedures. It explains how convertible bonds and some types of path-dependent derivatives can be valued using trees. It discusses the special problems associated with valuing barrier options numerically and how these problems can be handled. Finally, it outlines alternative ways of constructing trees for two correlated variables and shows how Monte Carlo simulation can be used to value derivatives when there are early exercise opportunities.

As in earlier chapters, results are presented for derivatives dependent on an asset providing a yield at rate q. For an option on a stock index, q should be set equal to the dividend yield on the index; for an option on a currency, it should be set equal to the foreign risk-free rate; for an option on a futures contract, it should be set equal to the domestic risk-free rate.

27.1 ALTERNATIVES TO BLACK–SCHOLES–MERTON

The Black–Scholes–Merton model assumes that an asset's price changes continously in a way that produces a lognormal distribution for the price at any future time. There are many alternative processes that can be assumed. One possibility is to retain the property that the asset price changes continuously, but assume a process other than geometric Brownian motion. Another alternative is to overlay continuous asset price changes with jumps. Yet another alternative is to assume a process where all the asset price changes that take place are jumps. We will consider examples of all three types of processes in this section. In particular, we will consider the constant elasticity of variance model, Merton's mixed jump–diffusion model, and the variance-gamma model. All three models are implemented in DerivaGem. The types of processes we consider in this section are known collectively as *Levy processes*.[1]

The Constant Elasticity of Variance Model

One alternative to Black–Scholes–Merton is the *constant elasticity of variance* (CEV) model. This is a diffusion model where the risk-neutral process for a stock price S is

$$dS = (r - q)S\,dt + \sigma S^{\beta}\,dz$$

where r is the risk-free rate, q is the dividend yield, dz is a Wiener process, σ is a volatility parameter, and β is a positive constant.[2] The parameter σ is roughly equivalent to the volatility in the usual lognormal model multiplied by $S^{1-\beta}$.

When $\beta = 1$, the CEV model is the geometric Brownian motion model we have been using up to now. When $\beta < 1$, the volatility increases as the stock price decreases. This creates a probability distribution similar to that observed for equities with a heavy left tail and less heavy right tail (see Figure 20.4).[3] When $\beta > 1$, the volatility increases as the stock price increases. This creates a probability distribution with a heavy right tail and a less heavy left tail. This corresponds to a volatility smile where the implied volatility is an increasing function of the strike price. This type of volatility smile is sometimes observed for options on futures.

The valuation formulas for European call and put options under the CEV model are

$$c = S_0 e^{-qT}[1 - \chi^2(a, b + 2, c)] - Ke^{-rT}\chi^2(c, b, a)$$
$$p = Ke^{-rT}[1 - \chi^2(c, b, a)] - S_0 e^{-qT}\chi^2(a, b + 2, c)$$

when $0 < \beta < 1$, and

$$c = S_0 e^{-qT}[1 - \chi^2(c, -b, a)] - Ke^{-rT}\chi^2(a, 2 - b, c)$$
$$p = Ke^{-rT}[1 - \chi^2(a, 2 - b, c)] - S_0 e^{-qT}\chi^2(c, -b, a)$$

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[1] Roughly speaking, a Levy process is a continuous-time stochastic process with stationary independent increments.

[2] See J. C. Cox and S. A. Ross, "The Valuation of Options for Alternative Stochastic Processes," *Journal of Financial Economics*, 3 (March 1976): 145–66.

[3] The reason is as follows. As the stock price decreases, the volatility increases making even lower stock price more likely; when the stock price increases, the volatility decreases making higher stock prices less likely.

when $\beta > 1$, with

$$a = \frac{[Ke^{-(r-q)T}]^{2(1-\beta)}}{(1-\beta)^2 v}, \qquad b = \frac{1}{1-\beta}, \qquad c = \frac{S^{2(1-\beta)}}{(1-\beta)^2 v}$$

where

$$v = \frac{\sigma^2}{2(r-q)(\beta-1)}[e^{2(r-q)(\beta-1)T} - 1]$$

and $\chi^2(z, k, v)$ is the cumulative probability that a variable with a noncentral χ^2 distribution with noncentrality parameter v and k degrees of freedom is less than z. A procedure for computing $\chi^2(z, k, v)$ is provided in Technical Note 12 on the author's website: www-2.rotman.utoronto.ca/~hull/TechnicalNotes.

The CEV model is useful for valuing exotic equity options. The parameters of the model can be chosen to fit the prices of plain vanilla options as closely as possible by minimizing the sum of the squared differences between model prices and market prices.

Merton's Mixed Jump–Diffusion Model

Merton has suggested a model where jumps are combined with continuous changes.[4] Define:

λ: Average number of jumps per year

k: Average jump size, measured as a percentage of the asset price

The percentage jump size is assumed to be drawn from a probability distribution in the model.

The probability of a jump in time Δt is $\lambda \Delta t$. The average growth rate in the asset price from the jumps is therefore λk. The risk-neutral process for the asset price is

$$\frac{dS}{S} = (r - q - \lambda k)\, dt + \sigma\, dz + dp$$

where dz is a Wiener process, dp is the Poisson process generating the jumps, and σ is the volatility of the non-jump component of S. The processes dz and dp are assumed to be independent.

An important particular case of Merton's model is where the logarithm of one plus the size of the percentage jump is normal. Assume that the standard deviation of the normal distribution is s. Merton shows that a European option price can then be written

$$\sum_{n=0}^{\infty} \frac{e^{-\lambda' T}(\lambda' T)^n}{n!} f_n$$

where $\lambda' = \lambda(1 + k)$. The variable f_n is the Black–Scholes–Merton option price when the dividend yield is q, the variance rate is

$$\sigma^2 + \frac{ns^2}{T}$$

[4] See R. C. Merton, "Option Pricing When Underlying Stock Returns Are Discontinuous," *Journal of Financial Economics*, 3 (March 1976): 125–44.

and the risk-free rate is

$$r - \lambda k + \frac{n\gamma}{T}$$

where $\gamma = \ln(1 + k)$.

This model gives rise to heavier left and heavier right tails than Black–Scholes–Merton. It can be used for pricing currency options. As in the case of the CEV model, the model parameters are chosen by minimizing the sum of the squared differences between model prices and market prices.

Simulating Jumps

Models that involve jumps can be implemented with Monte Carlo simulation. When jumps are generated by a Poisson process, the probability of exactly m jumps in time t is

$$\frac{e^{-\lambda t}(\lambda t)^m}{m!}$$

where λ is the average number of jumps per year. Equivalently, λt is the average number of jumps in time t.

Suppose that on average 0.5 jumps happen per year. The probability of m jumps in 2 years is

$$\frac{e^{-0.5\times 2}(0.5 \times 2)^m}{m!}$$

Table 27.1 gives the probability and cumulative probability of 0, 1, 2, 3, 4, 5, 6, 7, and 8 jumps in 2 years. (The numbers in a table such as this can be calculated using the POISSON function in Excel.)

To simulate a process following jumps over 2 years, it is necessary to determine on each simulation trial:

1. The number of jumps
2. The size of each jump.

To determine the number of jumps, on each simulation trial we sample a random number between 0 and 1 and use Table 27.1 as a look-up table. If the random number is between 0 and 0.3679, no jumps occur; if the random number is between 0.3679 and

Table 27.1 Probabilities for number of jumps in 2 years.

Number of jumps, m	*Probability of exactly m jumps*	*Probability of m jumps or less*
0	0.3679	0.3679
1	0.3679	0.7358
2	0.1839	0.9197
3	0.0613	0.9810
4	0.0153	0.9963
5	0.0031	0.9994
6	0.0005	0.9999
7	0.0001	1.0000
8	0.0000	1.0000

0.7358, one jump occurs; if the random number is between 0.7358 and 0.9197, two jumps occur; and so on. To determine the size of each jump, it is necessary on each simulation trial to sample from the probability distribution for the jump size once for each jump that occurs. Once the number of jumps and the jump sizes have been determined, the final value of the variable being simulated is known for the simulation trial.

In a mixed jump–diffusion model, jumps are superimposed upon the usual lognormal diffusion process that is assumed for stock prices. The process then has two components (the usual diffusion component and the jump component) and each must be sampled separately. The diffusion component is sampled as described in Sections 21.6 and 21.7 while the jump component is sampled as just described. When derivatives are valued, it is important to ensure that the overall expected return from the asset (from both components) is the risk-free rate.

The Variance-Gamma Model

An example of a pure jump model that is proving quite popular is the *variance-gamma model*.[5] Define a variable g as the change over time T in a variable that follows a gamma process with mean rate of 1 and variance rate of v. A gamma process is a pure jump process where small jumps occur very frequently and large jumps occur only occasionally. The probability density for g is

$$\frac{g^{T/v-1}e^{-g/v}}{v^{T/v}\Gamma(T/v)}$$

where $\Gamma(\cdot)$ denotes the gamma function. This probability density can be computed in Excel using the GAMMADIST$(\cdot, \cdot, \cdot, \cdot)$ function. The first argument of the function is g, the second is T/v, the third is v, and the fourth is TRUE or FALSE, where TRUE returns the cumulative probability distribution function and FALSE returns the probability density function we have just given.

As usual, we define S_T as the asset price at time T, S_0 as the asset price today, r as the risk-free interest rate, and q as the dividend yield. Conditional on g, $\ln S_T$ has a risk-neutral probability distribution under the variance-gamma model that is normal. The conditional mean is

$$\ln S_0 + (r - q)T + \omega + \theta g$$

and the conditional standard deviation is

$$\sigma\sqrt{g}$$

where

$$\omega = (T/v)\ln(1 - \theta v - \sigma^2 v/2)$$

The variance-gamma model has three parameters: v, σ, and θ.[6] The parameter v is the variance rate of the gamma process, σ is the volatility, and θ is a parameter defining skewness. When $\theta = 0$, $\ln S_T$ is symmetric; when $\theta < 0$, it is negatively skewed (as for equities); and when $\theta > 0$, it is positively skewed.

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[5] See D. B. Madan, P. P. Carr, and E. C. Chang, "The Variance-Gamma Process and Option Pricing," *European Finance Review*, 2 (1998): 79–105.

[6] Note that all these parameters are liable to change when we move from the real world to the risk-neutral world. This is in contrast to pure diffusion models where the volatility remains the same.

Suppose that we are interested in using Excel to obtain 10,000 random samples of the change in an asset price between time 0 and time T using the variance-gamma model. As a preliminary, we could set cells E1, E2, E3, E4, E5, E6, and E7 equal to T, ν, θ, σ, r, q, and S_0, respectively. We could also set E8 equal to ω by defining it as

$$= \$E\$1 * LN(1 - \$E\$3 * \$E\$2 - \$E\$4 * \$E\$4 * \$E\$2/2)/\$E\$2$$

We could then proceed as follows:

1. Sample values for g using the GAMMAINV function. Set the contents of cells A1, A2, . . . , A10000 as

$$= GAMMAINV(RAND(), \$E\$1/\$E\$2, \$E\$2)$$

2. For each value of g we sample a value z for a variable that is normally distributed with mean θg and standard deviation $\sigma\sqrt{g}$. This can be done by defining cell B1 as

$$= A1 * \$E\$3 + SQRT(A1) * \$E\$4 * NORMSINV(RAND())$$

   and cells B2, B3, . . . , B10000 similarly.

3. The stock price S_T is given by

$$S_T = S_0 \exp[(r - q)T + \omega + z]$$

   By defining C1 as

$$= \$E\$7 * EXP((\$E\$5 - \$E\$6) * \$E\$1 + B1 + \$E\$8)$$

   and C2, C3, . . . , C10000 similarly, random samples from the distribution of S_T are created in these cells.

Figure 27.1 Distributions obtained with variance-gamma process and geometric Brownian motion.

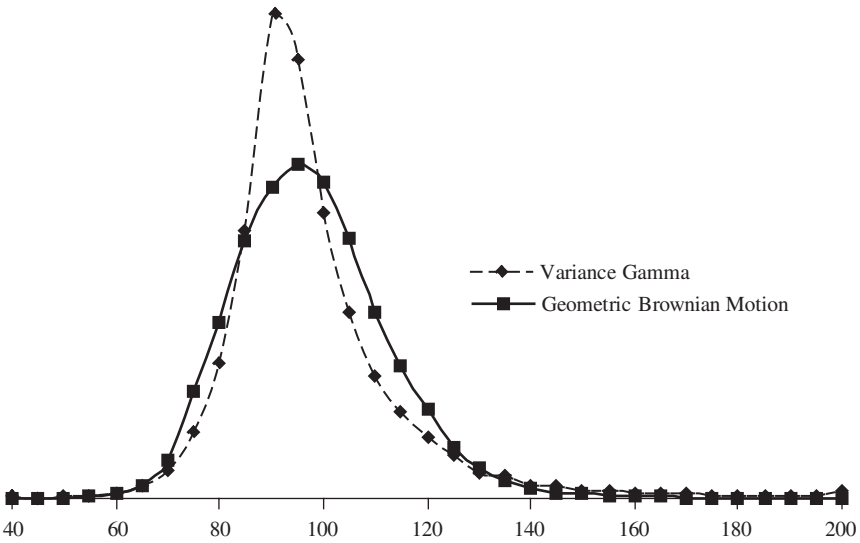


Figure 27.1 shows the probability distribution that is obtained using the variance-gamma model for S_T when $S_0 = 100$, $T = 0.5$, $v = 0.5$, $\theta = 0.1$, $\sigma = 0.2$, and $r = q = 0$. For comparison it also shows the distribution given by geometric Brownian motion when the volatility, σ is 0.2 (or 20%). Although not clear in Figure 27.1, the variance-gamma distribution has heavier tails than the lognormal distribution given by geometric Brownian motion.

One way of characterizing the variance-gamma distribution is that g defines the rate at which information arrives during time T. If g is large, a great deal of information arrives and the sample we take from a normal distribution in step 2 above has a relatively large mean and variance. If g is small, relatively little information arrives and the sample we take has a relatively small mean and variance. The parameter T is the usual time measure, and g is sometimes referred to as measuring economic time or time adjusted for the flow of information.

Semi-analytic European option valuation formulas are provided by Madan *et al.* (1998). The variance-gamma model tends to produce a U-shaped volatility smile. The smile is not necessarily symmetrical. It is very pronounced for short maturities and "dies away" for long maturities. The model can be fitted to either equity or foreign currency plain vanilla option prices.

27.2 STOCHASTIC VOLATILITY MODELS

The Black–Scholes–Merton model assumes that volatility is constant. In practice, as discussed in Chapter 23, volatility varies through time. The variance-gamma model reflects this with its g variable. Low values of g correspond to a low arrival rate for information and a low volatility; high values of g correspond to a high arrival rate for information and a high volatility.

An alternative to the variance-gamma model is a model where the process followed by the volatility variable is specified explicitly. Suppose first that the volatility parameter in the geometric Brownian motion is a known function of time. The risk-neutral process followed by the asset price is then

$$dS = (r - q)S\,dt + \sigma(t)S\,dz \qquad \textbf{(27.1)}$$

The Black–Scholes–Merton formulas are then correct provided that the variance rate is set equal to the average variance rate during the life of the option (see Problem 27.6). The variance rate is the square of the volatility. Suppose that during a 1-year period the volatility of a stock will be 20% during the first 6 months and 30% during the second 6 months. The average variance rate is

$$0.5 \times 0.20^2 + 0.5 \times 0.30^2 = 0.065$$

It is correct to use Black–Scholes–Merton with a variance rate of 0.065. This corresponds to a volatility of $\sqrt{0.065} = 0.255$, or 25.5%.

Equation (27.1) assumes that the instantaneous volatility of an asset is perfectly predictable. In practice, volatility varies stochastically. This has led to the development of more complex models with two stochastic variables: the stock price and its volatility.

One model that has been used by researchers is

$$dS/S = (r - q)\,dt + \sqrt{V}\,dz_S \tag{27.2}$$

$$dV = a(V_L - V)\,dt + \xi V^{\alpha}\,dz_V \tag{27.3}$$

where a, V_L, ξ, and α are constants, and dz_S and dz_V are Wiener processes. The variable V in this model is the asset's variance rate. The variance rate has a drift that pulls it back to a level V_L at rate a.

Hull and White show that, when volatility is stochastic but uncorrelated with the asset price, the price of a European option is the Black–Scholes–Merton price integrated over the probability distribution of the average variance rate during the life of the option.[7] Thus, a European call price is

$$\int_0^{\infty} c(\overline{V})\, g(\overline{V})\, d\overline{V}$$

where $\overline{V}$ is the average value of the variance rate, c is the Black–Scholes–Merton price expressed as a function of $\overline{V}$, and g is the probability density function of $\overline{V}$ in a risk-neutral world. This result can be used to show that Black–Scholes–Merton overprices options that are at the money or close to the money, and underprices options that are deep-in-or deep-out-of-the-money. The model is consistent with the pattern of implied volatilities observed for currency options (see Section 20.2).

The case where the asset price and volatility are correlated is more complicated. Option prices can be obtained using Monte Carlo simulation. In the particular case where $\alpha = 0.5$, Hull and White provide a series expansion and Heston provides an analytic result.[8] The pattern of implied volatilities obtained when the volatility is negatively correlated with the asset price is similar to that observed for equities (see Section 20.3).[9]

Chapter 23 discusses exponentially weighted moving average (EWMA) and GARCH(1,1) models. These are alternative approaches to characterizing a stochastic volatility model. Duan shows that it is possible to use GARCH(1,1) as the basis for an internally consistent option pricing model.[10] (See Problem 23.14 for the equivalence of GARCH(1,1) and stochastic volatility models.)

The SABR Model

A stochastic volatility model that has become very popular with practitioners for valuing European options (particularly interest rate options) maturing at a particular time T is the SABR model developed by Hagan *et al.* (2002).[11] The advantage of the

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[7] See J. C. Hull and A. White, "The Pricing of Options on Assets with Stochastic Volatilities," *Journal of Finance*, 42 (June 1987): 281–300. This result is independent of the process followed by the variance rate.

[8] See J. C. Hull and A. White, "An Analysis of the Bias in Option Pricing Caused by a Stochastic Volatility," *Advances in Futures and Options Research*, 3 (1988): 27–61; S. L. Heston, "A Closed Form Solution for Options with Stochastic Volatility with Applications to Bonds and Currency Options," *Review of Financial Studies*, 6, 2 (1993): 327–43. The Heston model is implemented in DerivaGem 4.00.

[9] The reason is given in footnote 3.

[10] See J.-C. Duan, "The GARCH Option Pricing Model," *Mathematical Finance*, vol. 5 (1995), 13–32; and J.-C. Duan, "Cracking the Smile" *RISK*, vol. 9 (December 1996), 55-59.

[11] See P. Hagan, D. Kumar, A. Lesniewski, and D. Woodward, "Managing Smile Risk," *Wilmott*, September 2002: 84–108, and also www.math.ku.dk/~rolf/SABR.pdf. SABR is short for "stochastic alpha, beta, and rho." We use σ instead of Hagan *et al.*'s α.

model is that it can fit the volatility smiles observed in practice reasonably well and is useful for managing risks associated with movements in the smile through time.

The model is

$$dF = \sigma F^{\beta}\, dz$$
$$\frac{d\sigma}{\sigma} = v\, dw$$

where F is a forward interest rate or the forward price of an asset, dz and dw are Wiener processes, σ is the stochastic volatility, and β and v are constants.[12] Three other parameters are ρ, the correlation between dz and dw, σ_0, the initial value of σ, and F_0, the initial value of F.

Hagan *et al.* derive an approximate formula for the Black-model implied volatility given by SABR for a European option with strike price K (see Sections 18.7 and 18.8 for Black's model). Define

$$x = (F_0K)^{(1-\beta)/2}, \qquad y = (1-\beta)\ln(F_0/K)$$

$$A = \frac{\sigma_0}{x(1 + y^2/24 + y^4/1920)}, \qquad B = 1 + \left(\frac{(1-\beta)^2\sigma_0^2}{24x^2} + \frac{\rho\beta v\sigma_0}{4x} + \frac{2-3\rho^2}{24}v^2\right)T$$

$$\phi = \frac{vx}{\sigma_0}\ln\left(\frac{F_0}{K}\right), \qquad \chi = \ln\left(\frac{\sqrt{1 - 2\rho\phi + \phi^2} + \phi - \rho}{1-\rho}\right)$$

The implied volatility is $AB\phi/\chi$. If $F_0 = K$, this becomes $\sigma_0 B/(F_0^{1-\beta})$.[13]

The parameter σ_0 defines the level of the volatility. (It is roughly equivalent to the volatility estimate in the usual lognormal model multiplied by $F_0^{1-\beta}$.) The parameter ρ defines the shape of the smile. For a large positive ρ, the smile is upward sloping; for a large negative ρ, it is downward sloping; and for intermediate values of ρ, it is U-shaped. As the parameter v increases, the smile become more accentuated. (See Problem 27.23.) The SABR model is implemented in DerivaGem 4.00.

Often β is chosen equal to 0.5 in interest rate applications of the SABR model. The parameters ρ, σ_0, and v that fit the smile typically depend on the maturity T being considered. However, as mentioned earlier, the model has proved a useful way of managing the risks in smile movements.

Rough Volatility Models

In Section 14.8, we introduced fractional Brownian motion. Gatheral, Jaisson, and Rosenbaum (2018) argue that fractional Brownian motion gives a better description of the behavior of volatility than regular Brownian motion.[14] Using high-frequency data, they find that a Hurst exponent between 0.06 and 0.20 fits the behavior of a range of different stock indices well. It also fits the volatility surfaces that are observed in practice

[12] This is the process for F in a world where F has zero drift. When interest rates are stochastic, it is correct to discount the expected payoff from the derivative in this world from time T to time zero at the T-year rate. This is explained in the next chapter.

[13] Other researchers have come up with slightly more accurate formulas, but this is the one that is most widely used.

[14] See J. Gatheral, T. Jaisson, and M. Rosenbaum, "Volatility is Rough," *Quantitative Finance*, 18, 6 (2018): 933–949.

better than simpler stochastic volatility models. Models which are based on fractional Brownian motion are known as *rough volatility models*.

One well-known rough volatility model is referred to as "rough Heston." It has the advantage of being analytically tractable. The original Heston model is given by equations (27.2) and (27.3) with $\alpha = 0.5$ and can be written as

$$\frac{dS_t}{S_t} = (r - q)\,dt + \sqrt{V_t}(\rho\,dz_t + \sqrt{1-\rho^2}\,dw_t)$$

$$dV_t = a(V_L - V_t)\,dt + \xi\sqrt{V_t}\,dz_t$$

where dz_t and dw_t are uncorrelated Wiener processes and ρ is the correlation between the stock price and its volatility. In the rough Heston model the process followed by the variance rate involves fractional Brownian motion rather than regular Brownian motion.

To value exotic options it is often necessary use Monte Carlo simulation. As explained in Section 14.8, in fractional Brownian motion the change in a future time period must be simulated so that it has the right correlation with changes in all previous time periods. This is computationally very slow and has led Jaber (2019) to suggest the lifted Heston model, which imitates rough Heston using only regular Brownian motion.[15]

27.3 THE IVF MODEL

The parameters of the models we have discussed so far can be chosen so that they provide an approximate fit to the prices of plain vanilla options on any given day. Financial institutions sometimes want to go one stage further and use a model that provides an exact fit to the prices of these options.[16] In 1994 Derman and Kani, Dupire, and Rubinstein developed a model that is designed to do this. It has become known as the *implied volatility function* (IVF) model or the *local volatility* model.[17] It provides an exact fit to the European option prices observed on any given day, regardless of the shape of the volatility surface.

The risk-neutral process for the asset price in the model has the form

$$dS = [r(t) - q(t)]S\,dt + \sigma(S,t)S\,dz$$

where $r(t)$ is the instantaneous forward interest rate for a contract maturing at time t and $q(t)$ is the dividend yield as a function of time. The volatility $\sigma(S,t)$ is a function of both S and t and is chosen so that the model prices all European options consistently

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[15] See, for example, O. El Euch, J. Gatheral, and M. Rosenbaum, "Roughening Heston," *Risk*, May (2019): 84–89, for a discussion of the rough volatility model, and E. A. Jaber, "Lifting the Heston Model," *Quantitative Finance*, 19 (2019): 1995–2013, for the lifted Heston model.

[16] There is a practical reason for this. If the bank does not use a model with this property, there is a danger that traders working for the bank will spend their time arbitraging the bank's internal models.

[17] See B. Dupire, "Pricing with a Smile," *Risk*, February (1994): 18–20; E. Derman and I. Kani, "Riding on a Smile," *Risk*, February (1994): 32–39; M. Rubinstein, "Implied Binomial Trees" *Journal of Finance*, 49, 3 (July 1994), 771–818.

with the market. It is shown both by Dupire and by Andersen and Brotherton-Ratcliffe that $\sigma(S, t)$ can be calculated analytically:[18]

$$[\sigma(K, T)]^2 = 2\frac{\partial c_{\text{mkt}}/\partial T + q(T)c_{\text{mkt}} + K[r(T) - q(T)]\partial c_{\text{mkt}}/\partial K}{K^2(\partial^2 c_{\text{mkt}}/\partial K^2)} \tag{27.4}$$

where $c_{\text{mkt}}(K, T)$ is the market price of a European call option with strike price K and maturity T. If a sufficiently large number of European call prices are available in the market, this equation can be used to estimate the $\sigma(S, t)$ function.[19]

Andersen and Brotherton-Ratcliffe implement the model by using equation (27.4) together with the implicit finite difference method. An alternative approach, the *implied tree* methodology suggested by Derman and Kani and Rubinstein, involves constructing a tree for the asset price that is consistent with option prices in the market.

When it is used in practice the IVF model is recalibrated daily to the prices of plain vanilla options. It is a tool to price exotic options consistently with plain vanilla options. As discussed in Chapter 20 plain vanilla options define the risk-neutral probability distribution of the asset price at all future times. It follows that the IVF model gets the risk-neutral probability distribution of the asset price at all future times correct. This means that options providing payoffs at just one time (e.g., cash-or-nothing and asset-or-nothing options) are priced correctly by the IVF model. However, the model does not necessarily get the joint distribution of the asset price at two or more times correct. This means that exotic options such as compound options and barrier options may be priced incorrectly.[20]

27.4 CONVERTIBLE BONDS

We now move on to discuss how the numerical procedures presented in Chapter 21 can be modified to handle particular valuation problems. We start by considering convertible bonds.

Convertible bonds are bonds issued by a company where the holder has the option to exchange the bonds for the company's stock at certain times in the future. The *conversion ratio* is the number of shares of stock obtained for one bond (this can be a function of time). The bonds are almost always callable (i.e., the issuer has the right to buy them back at certain times at a predetermined prices). The holder always has the right to convert the bond once it has been called. The call feature is therefore usually a way of forcing

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[18] See B. Dupire, "Pricing with a Smile," *Risk*, February (1994), 18–20; L. B. G. Andersen and R. Brotherton-Ratcliffe "The Equity Option Volatility Smile: An Implicit Finite Difference Approach," *Journal of Computation Finance* 1, No. 2 (Winter 1997/98): 5–37. Dupire considers the case where r and q are zero; Andersen and Brotherton-Ratcliffe consider the more general situation.

[19] Some smoothing of the observed volatility surface is typically necessary.

[20] Hull and Suo test the IVF model by assuming that all derivative prices are determined by a stochastic volatility model. They found that the model works reasonably well for compound options, but sometimes gives serious errors for barrier options. See J. C. Hull and W. Suo, "A Methodology for the Assessment of Model Risk and its Application to the Implied Volatility Function Model," *Journal of Financial and Quantitative Analysis*, 37, 2 (June 2002): 297–318

conversion earlier than the holder would otherwise choose. Sometimes the holder's call option is conditional on the price of the company's stock being above a certain level.

Credit risk plays an important role in the valuation of convertibles. If credit risk is ignored, poor prices are obtained because the coupons and principal payments on the bond are overvalued. Ingersoll provides a way of valuing convertibles using a model similar to Merton's (1974) model discussed in Section 24.6.[21] He assumes geometric Brownian motion for the issuer's total assets and models the company's equity, its convertible debt, and its other debt as claims contingent on the value of the assets. Credit risk is taken into account because the debt holders get repaid in full only if the value of the assets exceeds the amount owing to them.

A simpler model that is widely used in practice involves modeling the issuer's stock price. It is assumed that the stock follows geometric Brownian motion except that there is a probability $\lambda \Delta t$ that there will be a default in each short period of time Δt. In the event of a default the stock price falls to zero and there is a recovery on the bond. The variable λ is the risk-neutral hazard rate defined in Section 24.2.

The stock price process can be represented by varying the usual binomial tree so that at each node there is:

1. A probability p_u of a percentage up movement of size u over the next time period of length Δt
2. A probability p_d of a percentage down movement of size d over the next time period of length Δt
3. A probability $\lambda \Delta t$, or more accurately $1 - e^{-\lambda \Delta t}$, that there will be a default with the stock price moving to zero over the next time period of length Δt

Assume that σ is the volatility of the stock price conditional on no default and p_{def} is the probability of default. Parameter values that match σ and give an overall expected return for the stock of $r - q$ are:

$$p_u = \frac{a - de^{-\lambda \Delta t}}{u - d}, \quad p_d = \frac{ue^{-\lambda \Delta t} - a}{u - d}, \quad p_{\text{def}} = 1 - e^{-\lambda \Delta t}, \quad u = e^{\sigma \sqrt{\Delta t}}, \quad d = \frac{1}{u}$$

where $a = e^{(r-q)\Delta t}$, r is the risk-free rate, and q is the dividend yield on the stock.

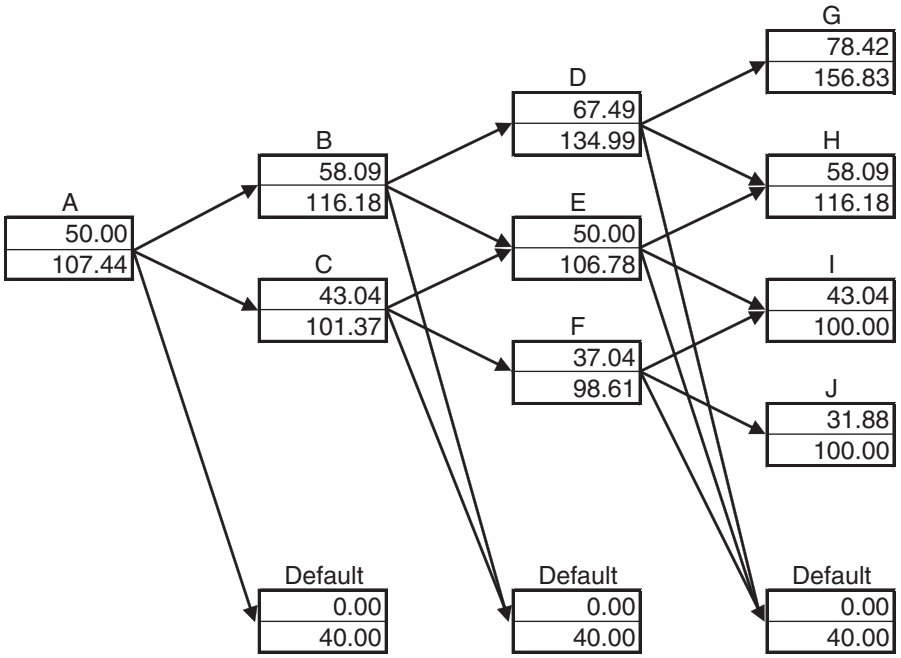
The life of the tree is set equal to the life of the convertible bond. The value of the convertible at the final nodes of the tree is calculated based on any conversion options that the holder has at that time. We then roll back through the tree. At nodes where the terms of the instrument allow conversion we test whether conversion is optimal. We also test whether the position of the issuer can be improved by calling the bonds. If so, we assume that the bonds are called and retest whether conversion is optimal.

Example 27.1

Consider a 9-month zero-coupon bond issued by company XYZ with a face value of \$100. Suppose that it can be exchanged for two shares of company XYZ's stock at any time during the 9 months. Assume also that it is callable for \$113 at any time. The initial stock price is \$50, its volatility is 30% per annum, and there are no dividends. The hazard rate λ is 1% per year, and all risk-free rates for all

[21] See J. E. Ingersoll, "A Contingent Claims Valuation of Convertible Securities," *Journal of Financial Economics*, 4, (May 1977), 289–322.

Figure 27.2 Tree for valuing convertible. Upper number at each node is stock price; lower number is convertible bond price.



maturities are 5%. Suppose that in the event of a default the bond is worth \$40 (i.e., the recovery rate, as it is usually defined, is 40%).

Figure 27.2 shows the stock price tree that can be used to value the convertible when there are three time steps ($\Delta t = 0.25$). The upper number at each node is the stock price; the lower number is the price of the convertible bond. The tree parameters are:

$$u = e^{0.3\sqrt{0.25}} = 1.1618, \quad d = 1/u = 0.8607$$

$$a = e^{0.05\times 0.25} = 1.0126, \quad p_u = 0.5115, \quad p_d = 0.4860, \quad p_{\text{def}} = 0.0025$$

At the three default nodes the stock price is zero and the bond price is 40.

Consider first the final nodes. At nodes G and H the bond should be converted and is worth twice the stock price. At nodes I and J the bond should not be converted and is worth 100.

Moving back through the tree enables the value to be calculated at earlier nodes. Consider, for example, node E. The value, if the bond is converted, is $2 \times 50 = \$100$. If it is not converted, then there is (a) a probability 0.5115 that it will move to node H, where the bond is worth 116.18, (b) a 0.4860 probability that it will move down to node I, where the bond is worth 100, and (c) a 0.0025 probability that it will default and be worth 40. The value of the bond if it is not converted is therefore

$$(0.5115 \times 116.18 + 0.4860 \times 100 + 0.0025 \times 40) \times e^{-0.05\times 0.25} = 106.78$$

This is more than the value of 100 that it would have if converted. We deduce that it is not worth converting the bond at node E. Finally, we note that the bond issuer would not call the bond at node E because this would be offering 113 for a bond worth 106.78.

As another example consider node B. The value of the bond if it is converted is $2 \times 58.09 = 116.18$. If it is not converted a similar calculation to that just given for node E gives its value as 119.54. The convertible bond holder will therefore choose not to convert. However, at this stage the bond issuer will call the bond for 113 and the bond holder will then decide that converting is better than being called. The value of the bond at node B is therefore 116.18. A similar argument is used to arrive at the value at node D. With no conversion the value is 135.08. However, the bond is called, forcing conversion and reducing the value at the node to 134.99.

The value of the convertible is its value at the initial node A, or 107.44.

When interest is paid on the debt, it must be taken into account. At each node, when valuing the bond on the assumption that it is not converted, the present value of any interest payable on the bond in the next time step should be included. The risk-neutral hazard rate λ can be estimated from either bond prices or credit default swap spreads. In a more general implementation, λ, σ, and r are functions of time. This can be handled using a trinomial rather than a binomial tree (see Section 21.4).

One disadvantage of the model we have presented is that the probability of default is independent of the stock price. This has led some researchers to suggest an implicit finite difference method implementation of the model where the hazard rate λ is a function of the stock price as well as time.[22]

27.5 PATH-DEPENDENT DERIVATIVES

A path-dependent derivative (or history-dependent derivative) is a derivative where the payoff depends on the path followed by the price of the underlying asset, not just its final value. Asian options and lookback options are examples of path-dependent derivatives. As explained in Chapter 26, the payoff from an Asian option depends on the average price of the underlying asset; the payoff from a lookback option depends on its maximum or minimum price. One approach to valuing path-dependent options when analytic results are not available is Monte Carlo simulation, as discussed in Chapter 21. A sample value of the derivative can be calculated by sampling a random path for the underlying asset in a risk-neutral world, calculating the payoff, and discounting the payoff at the risk-free interest rate. An estimate of the value of the derivative is found by obtaining many sample values of the derivative in this way and calculating their mean.

The main problem with Monte Carlo simulation is that the computation time necessary to achieve the required level of accuracy can be unacceptably high. Also, American-style path-dependent derivatives (i.e., path-dependent derivatives where one side has exercise opportunities or other decisions to make) cannot easily be handled. In

[22] See, e.g., L. B. G. Andersen and D. Buffum, "Calibration and Implementation of Convertible Bond Models," *Journal of Computational Finance*, 7, 1 (Winter 2003/04), 1–34. These authors suggest assuming that the hazard rate is inversely proportional to S^{α}, where S is the stock price and α is a positive constant.

this section, we show how the binomial tree methods presented in Chapter 21 can be extended to cope with some path-dependent derivatives.[23] The procedure can handle American-style path-dependent derivatives and is computationally more efficient than Monte Carlo simulation for European-style path-dependent derivatives.

For the procedure to work, two conditions must be satisfied:

1. The payoff from the derivative must depend on a single function, F, of the path followed by the underlying asset.
2. It must be possible to calculate the value of F at time $\tau + \Delta t$ from the value of F at time τ and the value of the underlying asset at time $\tau + \Delta t$.

First we construct a tree for the underlying asset's price in the usual way. The next step is to work forward through the tree establishing the maximum and minimum values of the path function at each node. Because the value of the path function at time $\tau + \Delta t$ depends only on the value of the path function at time τ and the value of the underlying variable at time $\tau + \Delta t$, the maximum and minimum values of the path function for the nodes at time $\tau + \Delta t$ can be calculated in a straightforward way from those for the nodes at time τ. The second stage is to choose representative values of the path function at each node. There are a number of approaches. A simple rule is to choose the representative values as the maximum value, the minimum value, and a number of other values that are equally spaced between the maximum and the minimum. As we roll back through the tree, we value the derivative for each of the representative values of the path function. When the value of the derivative is needed for other values of the path function, it is calculated by interpolation.

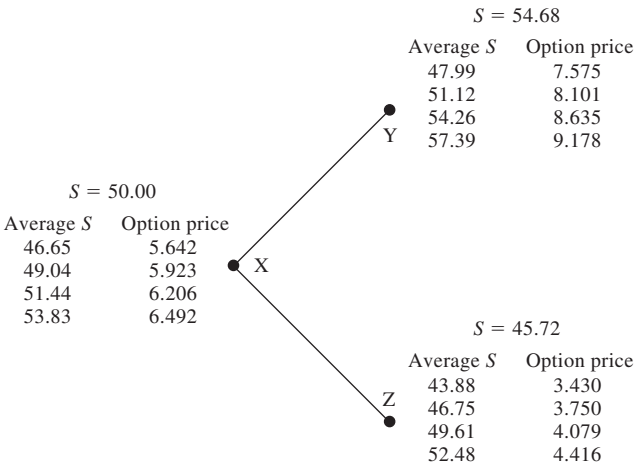
To illustrate the nature of the calculation, consider the problem of valuing the average price call option in Example 26.3 of Section 26.13 when the payoff depends on the arithmetic average stock price. The initial stock price is 50, the strike price is 50, the risk-free interest rate is 10%, the stock price volatility is 40%, and the time to maturity is 1 year. For 20 time steps, the binomial tree parameters are $\Delta t = 0.05$, $u = 1.0936$, $d = 0.9144$, $p = 0.5056$, and $1 - p = 0.4944$. The path function is the arithmetic average of the stock price.

Figure 27.3 shows the calculations that are carried out in one small part of the tree. Node X is the central node at time 0.2 year (at the end of the fourth time step). Nodes Y and Z are the two nodes at time 0.25 year that are reachable from node X. The stock price at node X is 50. Forward induction shows that the maximum average stock price that is achievable in reaching node X is 53.83. The minimum is 46.65. (The initial and final stock prices are included when calculating the average.) From node X, the tree branches to one of the two nodes Y and Z. At node Y, the stock price is 54.68 and the bounds for the average are 47.99 and 57.39. At node Z, the stock price is 45.72 and the bounds for the average stock price are 43.88 and 52.48.

Suppose that the representative values of the average are chosen to be four equally spaced values at each node. This means that, at node X, averages of 46.65, 49.04, 51.44, and 53.83 are considered; at node Y, averages 47.99, 51.12, 54.26, and 57.39 are considered; and at node Z, averages 43.88, 46.75, 49.61, and 52.48 are considered. Assume that backward induction has already been used to calculate the value of the option for each of the alternative values of the average at nodes Y and Z. Values are

[23] This approach was suggested in J. C. Hull and A. White, "Efficient Procedures for Valuing European and American Path-Dependent Options," *Journal of Derivatives*, 1, 1 (Fall 1993): 21–31.

Figure 27.3 Part of tree for valuing option on the arithmetic average.



shown in Figure 27.3 (e.g., at node Y when the average is 51.12, the value of the option is 8.101).

Consider the calculations at node X for the case where the average is 51.44. If the stock price moves up to node Y, the new average will be

$$\frac{5 \times 51.44 + 54.68}{6} = 51.98$$

The value of the derivative at node Y for this average can be found by interpolating between the values when the average is 51.12 and when it is 54.26. It is

$$\frac{(51.98 - 51.12) \times 8.635 + (54.26 - 51.98) \times 8.101}{54.26 - 51.12} = 8.247$$

Similarly, if the stock price moves down to node Z, the new average will be

$$\frac{5 \times 51.44 + 45.72}{6} = 50.49$$

and by interpolation the value of the derivative is 4.182.

The value of the derivative at node X when the average is 51.44 is, therefore,

$$(0.5056 \times 8.247 + 0.4944 \times 4.182)e^{-0.1 \times 0.05} = 6.206$$

The other values at node X are calculated similarly. Once the values at all nodes at time 0.2 year have been calculated, the nodes at time 0.15 year can be considered.

The value given by the full tree for the option at time zero is 7.17. As the number of time steps and the number of averages considered at each node is increased, the value of the option converges to the correct answer. With 60 time steps and 100 averages at each node, the value of the option is 5.58. The analytic approximation for the value of the option, as calculated in Example 26.3, with continuous averaging is 5.62.

A key advantage of the method described here is that it can handle American options. The calculations are as we have described them except that we test for early exercise at each node for each of the alternative values of the path function at the node. (In practice, the early exercise decision is liable to depend on both the value of the path function and the value of the underlying asset.) Consider the American version of the average price call considered here. The value calculated using the 20-step tree and four averages at each node is 7.77; with 60 time steps and 100 averages, the value is 6.17.

The approach just described can be used in a range of different situations. The two conditions that must be satisfied were listed at the beginning of this section. Efficiency is improved somewhat if quadratic rather than linear interpolation is used at each node.

27.6 BARRIER OPTIONS

Chapter 26 presented analytic results for standard barrier options. This section considers numerical procedures that can be used for barrier options when there are no analytic results.

In principle, many barrier options can be valued using the binomial and trinomial trees discussed in Chapter 21. Consider an up-and-out option. A simple approach is to value this in the same way as a regular option except that, when a node above the barrier is encountered, the value of the option is set equal to zero.

Trinomial trees work better than binomial trees, but even for them convergence is very slow when the simple approach is used. A large number of time steps are required to obtain a reasonably accurate result. The reason for this is that the barrier being assumed by the tree is different from the true barrier.[24] Define the *inner barrier* as the barrier formed by nodes just on the inside of the true barrier (i.e., closer to the center of the tree) and *the outer barrier* as the barrier formed by nodes just outside the true barrier (i.e., farther away from the center of the tree). Figure 27.4 shows the inner and outer barrier for a trinomial tree on the assumption that the true barrier is horizontal. The usual tree calculations implicitly assume that the outer barrier is the true barrier because the barrier conditions are first used at nodes on this barrier. When the time step is Δt, the vertical spacing between the nodes is of order $\sqrt{\Delta t}$. This means that errors created by the difference between the true barrier and the outer barrier also tend to be of order $\sqrt{\Delta t}$.

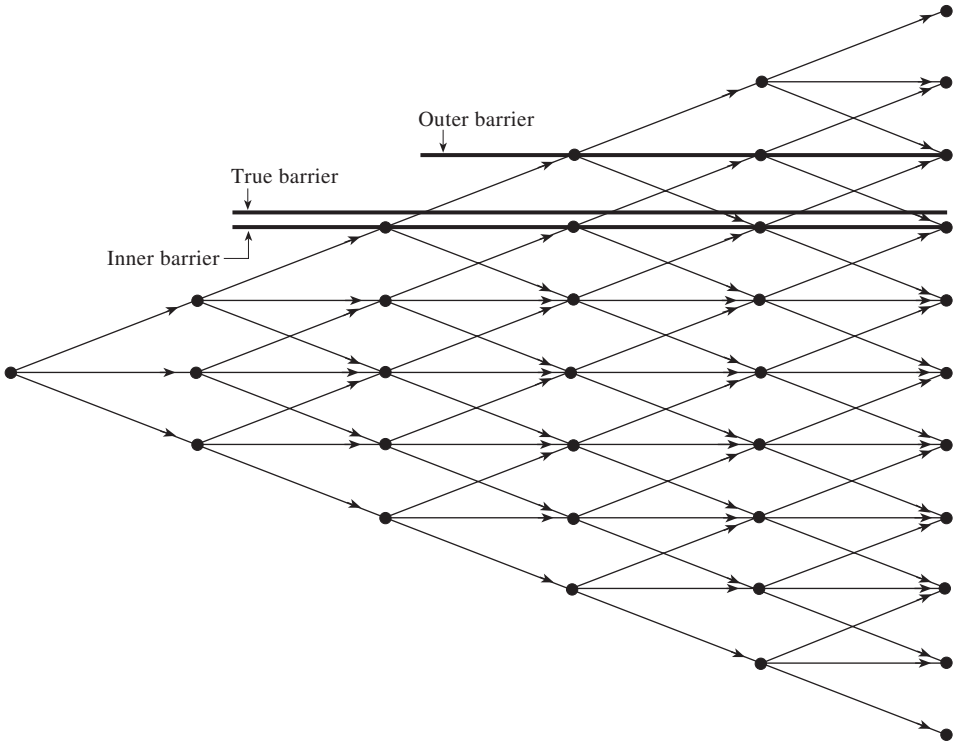
One approach to overcoming this problem is to:

1. Calculate the price of the derivative on the assumption that the inner barrier is the true barrier.
2. Calculate the value of the derivative on the assumption that the outer barrier is the true barrier.
3. Interpolate between the two prices.

Another approach is to ensure that nodes lie on the barrier. Suppose that the initial stock price is S_0 and that the barrier is at H. In a trinomial tree, there are three possible

[24] For a discussion of this, see P. P. Boyle and S. H. Lau, "Bumping Up Against the Barrier with the Binomial Method," *Journal of Derivatives*, 1, 4 (Summer 1994): 6–14.

Figure 27.4 Barriers assumed by trinomial trees.



movements in the asset's price at each node: up by a proportional amount u; stay the same; and down by a proportional amount d, where $d = 1/u$. We can always choose u so that nodes lie on the barrier. The condition that must be satisfied by u is

$$H = S_0 u^N$$

or

$$\ln H = \ln S_0 + N \ln u$$

for some positive or negative N.

When discussing trinomial trees in Section 21.4, the value suggested for u was $e^{\sigma\sqrt{3\Delta t}}$, so that $\ln u = \sigma\sqrt{3\Delta t}$. In the situation considered here, a good rule is to choose $\ln u$ as close as possible to this value, consistent with the condition given above. This means that

$$\ln u = \frac{\ln H - \ln S_0}{N}$$

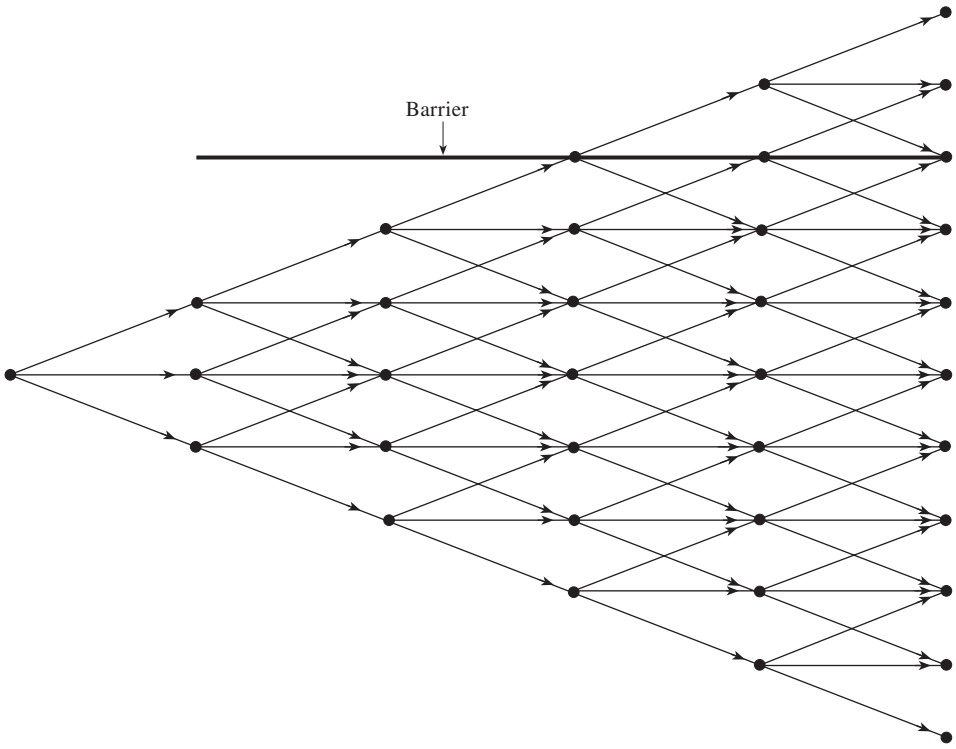
where

$$N = \text{int}\left[\frac{\ln H - \ln S_0}{\sigma\sqrt{3\Delta t}} + 0.5\right]$$

and $\text{int}(x)$ is the integral part of x.

This leads to a tree of the form shown in Figure 27.5. The probabilities p_u, p_m, and p_d on the upper, middle, and lower branches of the tree are chosen to match the first

Figure 27.5 Tree with nodes lying on barrier.



two moments of the return, so that

$$p_d = -\frac{(r - q - \sigma^2/2)\Delta t}{2 \ln u} + \frac{\sigma^2 \Delta t}{2(\ln u)^2},\ p_m = 1 - \frac{\sigma^2 \Delta t}{(\ln u)^2},\ p_u = \frac{(r - q - \sigma^2/2)\Delta t}{2 \ln u} + \frac{\sigma^2 \Delta t}{2(\ln u)^2}$$

The methods presented so far work reasonably well when the initial asset price is not close to the barrier. When the initial asset price is close to a barrier, the adaptive mesh model, which was introduced in Section 21.4, can be used.[25] The idea behind the model is that computational efficiency can be improved by grafting a fine tree onto a coarse tree to achieve a more detailed modeling of the asset price in the regions of the tree where it is needed most. To value a barrier option, it is computationally efficient to have a fine tree close to barriers with nodes on the barriers.

27.7 OPTIONS ON TWO CORRELATED ASSETS

Another tricky numerical problem is that of valuing American options dependent on two assets whose prices are correlated. A number of alternative approaches have been suggested. This section will explain three of these.

[25] See S. Figlewski and B. Gao, "The Adaptive Mesh Model: A New Approach to Efficient Option Pricing," *Journal of Financial Economics*, 53 (1999): 313–51.

Transforming Variables

It is relatively easy to construct a tree in three dimensions to represent the movements of two *uncorrelated* variables. The procedure is as follows. First, construct a two-dimensional tree for each variable, and then combine these trees into a single three-dimensional tree. The probabilities on the branches of the three-dimensional tree are the product of the corresponding probabilities on the two-dimensional trees. Suppose, for example, that the variables are stock prices, S_1 and S_2. Each can be represented in two dimensions by a Cox, Ross, and Rubinstein binomial tree. Assume that S_1 has a probability p_1 of moving up by a proportional amount u_1 and a probability $1 - p_1$ of moving down by a proportional amount d_1. Suppose further that S_2 has a probability p_2 of moving up by a proportional amount u_2 and a probability $1 - p_2$ of moving down by a proportional amount d_2. In the three-dimensional tree there are four branches emanating from each node. The probabilities are:

$p_1 p_2$: S_1 increases; S_2 increases
$p_1(1 - p_2)$: S_1 increases; S_2 decreases
$(1 - p_1)p_2$: S_1 decreases; S_2 increases
$(1 - p_1)(1 - p_2)$: S_1 decreases; S_2 decreases

Consider next the situation where S_1 and S_2 are correlated. Suppose that the risk-neutral processes are:

$$dS_1 = (r - q_1)S_1\,dt + \sigma_1 S_1\,dz_1$$
$$dS_2 = (r - q_2)S_2\,dt + \sigma_2 S_2\,dz_2$$

and the instantaneous correlation between the Wiener processes, dz_1 and dz_2, is ρ. This means that

$$d\ln S_1 = (r - q_1 - \sigma_1^2/2)\,dt + \sigma_1\,dz_1$$
$$d\ln S_2 = (r - q_2 - \sigma_2^2/2)\,dt + \sigma_2\,dz_2$$

Two new uncorrelated variables can be defined:[26]

$$x_1 = \sigma_2 \ln S_1 + \sigma_1 \ln S_2$$
$$x_2 = \sigma_2 \ln S_1 - \sigma_1 \ln S_2$$

These variables follow the processes

$$dx_1 = [\sigma_2(r - q_1 - \sigma_1^2/2) + \sigma_1(r - q_2 - \sigma_2^2/2)]\,dt + \sigma_1\sigma_2\sqrt{2(1 + \rho)}\,dz_A$$
$$dx_2 = [\sigma_2(r - q_1 - \sigma_1^2/2) - \sigma_1(r - q_2 - \sigma_2^2/2)]\,dt + \sigma_1\sigma_2\sqrt{2(1 - \rho)}\,dz_B$$

where dz_A and dz_B are uncorrelated Wiener processes.

The variables x_1 and x_2 can be modeled using two separate binomial trees. In time Δt, x_i has a probability p_i of increasing by h_i and a probability $1 - p_i$ of decreasing by h_i. The variables h_i and p_i are chosen so that the tree gives correct values for the first two moments of the distribution of x_1 and x_2. Because they are uncorrelated, the two trees can be combined into a single three-dimensional tree, as already described. At each

[26] This idea was suggested in J. C. Hull and A. White, "Valuing Derivative Securities Using the Explicit Finite Difference Method," *Journal of Financial and Quantitative Analysis*, 25 (1990): 87–100.

node of the tree, S_1 and S_2 can be calculated from x_1 and x_2 using the inverse relationships

$$S_1 = \exp\left[\frac{x_1 + x_2}{2\sigma_2}\right] \quad \text{and} \quad S_2 = \exp\left[\frac{x_1 - x_2}{2\sigma_1}\right]$$

The procedure for rolling back through a three-dimensional tree to value a derivative is analogous to that for a two-dimensional tree.

Using a Nonrectangular Tree

Rubinstein has suggested a way of building a three-dimensional tree for two correlated stock prices by using a nonrectangular arrangement of the nodes.[27] From a node (S_1, S_2), where the first stock price is S_1 and the second stock price is S_2, there is a 0.25 chance of moving to each of the following:

$$(S_1u_1, S_2A), \quad (S_1u_1, S_2B), \quad (S_1d_1, S_2C), \quad (S_1d_1, S_2D)$$

where

$$u_1 = \exp[(r - q_1 - \sigma_1^2/2)\,\Delta t + \sigma_1\sqrt{\Delta t}]$$
$$d_1 = \exp[(r - q_1 - \sigma_1^2/2)\,\Delta t - \sigma_1\sqrt{\Delta t}]$$

and

$$A = \exp[(r - q_2 - \sigma_2^2/2)\Delta t + \sigma_2\sqrt{\Delta t}(\rho + \sqrt{1 - \rho^2})]$$
$$B = \exp[(r - q_2 - \sigma_2^2/2)\Delta t + \sigma_2\sqrt{\Delta t}(\rho - \sqrt{1 - \rho^2})]$$
$$C = \exp[(r - q_2 - \sigma_2^2/2)\Delta t - \sigma_2\sqrt{\Delta t}(\rho - \sqrt{1 - \rho^2})]$$
$$D = \exp[(r - q_2 - \sigma_2^2/2)\Delta t - \sigma_2\sqrt{\Delta t}(\rho + \sqrt{1 - \rho^2})]$$

When the correlation is zero, this method is equivalent to constructing separate trees for S_1 and S_2 using the alternative binomial tree construction method in Section 21.4.

Adjusting the Probabilities

A third approach to building a three-dimensional tree for S_1 and S_2 involves first assuming no correlation and then adjusting the probabilities at each node to reflect the correlation.[28] The alternative binomial tree construction method for each of S_1 and S_2 in Section 21.4 is used. This method has the property that all probabilities are 0.5. When the two binomial trees are combined on the assumption that there is no correlation, the probabilities are as shown in Table 27.2. When the probabilities are adjusted to reflect the correlation, they become those shown in Table 27.3.

27.8 MONTE CARLO SIMULATION AND AMERICAN OPTIONS

Monte Carlo simulation is well suited to valuing path-dependent options and options where there are many stochastic variables. Trees and finite difference methods are well

[27] See M. Rubinstein, "Return to Oz," *Risk*, November (1994): 67–70.

[28] This approach was suggested in the context of interest rate trees in J. C. Hull and A. White, "Numerical Procedures for Implementing Term Structure Models II: Two-Factor Models," *Journal of Derivatives*, Winter (1994): 37–48.

Table 27.2 Combination of binomials assuming no correlation.

S_2-move	S_1-move	
	Down	*Up*
Up	0.25	0.25
Down	0.25	0.25

Table 27.3 Combination of binomials assuming correlation of ρ.

S_2-move	S_1-move	
	Down	*Up*
Up	$0.25(1 - \rho)$	$0.25(1 + \rho)$
Down	$0.25(1 + \rho)$	$0.25(1 - \rho)$

suited to valuing American-style options. What happens if an option is both path dependent and American? What happens if an American option depends on several stochastic variables? Section 27.5 explained a way in which the binomial tree approach can be modified to value path-dependent options in some situations. A number of researchers have adopted a different approach by searching for a way in which Monte Carlo simulation can be used to value American-style options.[29] This section explains two alternative ways of proceeding.

The Least-Squares Approach

In order to value an American-style option it is necessary to choose between exercising and continuing at each early exercise point. The value of exercising is normally easy to determine. A number of researchers including Longstaff and Schwartz provide a way of

Table 27.4 Sample paths for put option example.

Path	$t = 0$	$t = 1$	$t = 2$	$t = 3$
1	1.00	1.09	1.08	1.34
2	1.00	1.16	1.26	1.54
3	1.00	1.22	1.07	1.03
4	1.00	0.93	0.97	0.92
5	1.00	1.11	1.56	1.52
6	1.00	0.76	0.77	0.90
7	1.00	0.92	0.84	1.01
8	1.00	0.88	1.22	1.34

[29] Tilley was the first researcher to publish a solution to the problem. See J. A. Tilley, "Valuing American Options in a Path Simulation Model," *Transactions of the Society of Actuaries*, 45 (1993): 83–104.

Table 27.5 Cash flows if exercise only possible at 3-year point.

Path	$t = 1$	$t = 2$	$t = 3$
1	0.00	0.00	0.00
2	0.00	0.00	0.00
3	0.00	0.00	0.07
4	0.00	0.00	0.18
5	0.00	0.00	0.00
6	0.00	0.00	0.20
7	0.00	0.00	0.09
8	0.00	0.00	0.00

determining the value of continuing when Monte Carlo simulation is used.[30] Their approach involves using a least-squares analysis to determine the best-fit relationship between the value of continuing and the values of relevant variables at each time an early exercise decision has to be made. The approach is best illustrated with a numerical example. We use the one in the Longstaff–Schwartz paper.

Consider a 3-year American put option on a non-dividend-paying stock that can be exercised at the end of year 1, the end of year 2, and the end of year 3. The risk-free rate is 6% per annum (continuously compounded). The current stock price is 1.00 and the strike price is 1.10. Assume that the eight paths shown in Table 27.4 are sampled for the stock price. (This example is for illustration only; in practice many more paths would be sampled.) If the option can be exercised only at the 3-year point, it provides a cash flow equal to its intrinsic value at that point. This is shown in the last column of Table 27.5.

If the put option is in the money at the 2-year point, the option holder must decide whether to exercise. Table 27.4 shows that the option is in the money at the 2-year point for paths 1, 3, 4, 6, and 7. For these paths, we assume an approximate relationship:

$$V = a + bS + cS^2$$

where S is the stock price at the 2-year point and V is the value of continuing, discounted back to the 2-year point. Our five observations on S are: 1.08, 1.07, 0.97, 0.77, and 0.84. From Table 27.5 the corresponding values for V are: 0.00, $0.07e^{-0.06\times 1}$, $0.18e^{-0.06\times 1}$, $0.20e^{-0.06\times 1}$, and $0.09e^{-0.06\times 1}$. The values of a, b, and c that minimize

$$\sum_{i=1}^{5}(V_i - a - bS_i - cS_i^2)^2$$

where S_i and V_i are the ith observation on S and V, respectively, are $a = -1.070$, $b = 2.983$ and $c = -1.813$, so that the best-fit relationship is

$$V = -1.070 + 2.983S - 1.813S^2$$

This gives the value at the 2-year point of continuing for paths 1, 3, 4, 6, and 7 of 0.0369, 0.0461, 0.1176, 0.1520, and 0.1565, respectively. From Table 27.4 the value of exercising

[30] See F. A. Longstaff and E. S. Schwartz, "Valuing American Options by Simulation: A Simple Least-Squares Approach," *Review of Financial Studies*, 14, 1 (Spring 2001): 113–47.

Table 27.6 Cash flows if exercise only possible at 2- and 3-year point.

Path	$t = 1$	$t = 2$	$t = 3$
1	0.00	0.00	0.00
2	0.00	0.00	0.00
3	0.00	0.00	0.07
4	0.00	0.13	0.00
5	0.00	0.00	0.00
6	0.00	0.33	0.00
7	0.00	0.26	0.00
8	0.00	0.00	0.00

is 0.02, 0.03, 0.13, 0.33, and 0.26. This means that we should exercise at the 2-year point for paths 4, 6, and 7. Table 27.6 summarizes the cash flows assuming exercise at either the 2-year point or the 3-year point for the eight paths.

Consider next the paths that are in the money at the 1-year point. These are paths 1, 4, 6, 7, and 8. From Table 27.4 the values of S for the paths are 1.09, 0.93, 0.76, 0.92, and 0.88, respectively. From Table 27.6, the corresponding continuation values discounted back to $t = 1$ are 0.00, $0.13e^{-0.06\times 1}$, $0.33e^{-0.06\times 1}$, $0.26e^{-0.06\times 1}$, and 0.00, respectively. The least-squares relationship is

$$V = 2.038 - 3.335S + 1.356S^2$$

This gives the value of continuing at the 1-year point for paths 1, 4, 6, 7, 8 as 0.0139, 0.1092, 0.2866, 0.1175, and 0.1533, respectively. From Table 27.4 the value of exercising is 0.01, 0.17, 0.34, 0.18, and 0.22, respectively. This means that we should exercise at the 1-year point for paths 4, 6, 7, and 8. Table 27.7 summarizes the cash flows assuming that early exercise is possible at all three times. The value of the option is determined by discounting each cash flow back to time zero at the risk-free rate and calculating the mean of the results. It is

$$\tfrac{1}{8}(0.07e^{-0.06\times 3} + 0.17e^{-0.06\times 1} + 0.34e^{-0.06\times 1} + 0.18e^{-0.06\times 1} + 0.22e^{-0.06\times 1}) = 0.1144$$

Since this is greater than 0.10, it is not optimal to exercise the option immediately.

Table 27.7 Cash flows from option.

Path	$t = 1$	$t = 2$	$t = 3$
1	0.00	0.00	0.00
2	0.00	0.00	0.00
3	0.00	0.00	0.07
4	0.17	0.00	0.00
5	0.00	0.00	0.00
6	0.34	0.00	0.00
7	0.18	0.00	0.00
8	0.22	0.00	0.00

This method can be extended in a number of ways. If the option can be exercised at any time we can approximate its value by considering a large number of exercise points (just as a binomial tree does). The relationship between V and S can be assumed to be more complicated. For example we could assume that V is a cubic rather than a quadratic function of S. The method can be used where the early exercise decision depends on several state variables. A functional form for the relationship between V and the variables is assumed and the parameters are estimated using the least-squares approach, as in the example just considered.

The Exercise Boundary Parameterization Approach

A number of researchers, such as Andersen, have proposed an alternative approach where the early exercise boundary is parameterized and the optimal values of the parameters are determined iteratively by starting at the end of the life of the option and working backward.[31] To illustrate the approach, we continue with the put option example and assume that the eight paths shown in Table 27.4 have been sampled. In this case, the early exercise boundary at time t can be parameterized by a critical value of S, $S^*(t)$. If the asset price at time t is below $S^*(t)$ we exercise at time t; if it is above $S^*(t)$ we do not exercise at time t. The value of $S^*(3)$ is 1.10. If the stock price is above 1.10 when $t = 3$ (the end of the option's life) we do not exercise; if it is below 1.10 we exercise. We now consider the determination of $S^*(2)$.

Suppose that we choose a value of $S^*(2)$ less than 0.77. The option is not exercised at the 2-year point for any of the paths. The value of the option at the 2-year point for the eight paths is then 0.00, 0.00, $0.07e^{-0.06\times 1}$, $0.18e^{-0.06\times 1}$, 0.00, $0.20e^{-0.06\times 1}$, $0.09e^{-0.06\times 1}$, and 0.00, respectively. The average of these is 0.0636. Suppose next that $S^*(2) = 0.77$. The value of the option at the 2-year point for the eight paths is then 0.00, 0.00, $0.07e^{-0.06\times 1}$, $0.18e^{-0.06\times 1}$, 0.00, 0.33, $0.09e^{-0.06\times 1}$, and 0.00, respectively. The average of these is 0.0813. Similarly when $S^*(2)$ equals 0.84, 0.97, 1.07, and 1.08, the average value of the option at the 2-year point is 0.1032, 0.0982, 0.0938, and 0.0963, respectively. This analysis shows that the optimal value of $S^*(2)$ (i.e., the one that maximizes the average value of the option) is 0.84. (More precisely, it is optimal to choose $0.84 \le S^*(2) < 0.97$.) When we choose this optimal value for $S^*(2)$, the value of the option at the 2-year point for the eight paths is 0.00, 0.00, 0.0659, 0.1695, 0.00, 0.33, 0.26, and 0.00, respectively. The average value is 0.1032.

We now move on to calculate $S^*(1)$. If $S^*(1) < 0.76$ the option is not exercised at the 1-year point for any of the paths and the value at the option at the 1-year point is $0.1032e^{-0.06\times 1} = 0.0972$. If $S^*(1) = 0.76$, the value of the option for each of the eight paths at the 1-year point is 0.00, 0.00, $0.0659e^{-0.06\times 1}$, $0.1695e^{-0.06\times 1}$, 0.0, 0.34, $0.26e^{-0.06\times 1}$, and 0.00, respectively. The average value of the option is 0.1008. Similarly when $S^*(1)$ equals 0.88, 0.92, 0.93, and 1.09 the average value of the option is 0.1283, 0.1202, 0.1215, and 0.1228, respectively. The analysis therefore shows that the optimal value of $S^*(1)$ is 0.88. (More precisely, it is optimal to choose $0.88 \le S^*(1) < 0.92$.) The value of the option at time zero with no early exercise is $0.1283e^{-0.06\times 1} = 0.1208$. This is greater than the value of 0.10 obtained by exercising at time zero.

In practice, tens of thousands of simulations are carried out to determine the early exercise boundary in the way we have described. Once the early exercise boundary has

[31] See L. B. G. Andersen, "A Simple Approach to the Pricing of Bermudan Swaptions in the Multifactor LIBOR Market Model," *Journal of Computational Finance*, 3, 2 (Winter 2000): 1–32.

been obtained, the paths for the variables are discarded and a new Monte Carlo simulation using the early exercise boundary is carried out to value the option. Our American put option example is simple in that we know that the early exercise boundary at a time can be defined entirely in terms of the value of the stock price at that time. In more complicated situations it is necessary to make assumptions about how the early exercise boundary should be parameterized.

Upper Bounds

The two approaches we have outlined tend to underprice American-style options because they assume a suboptimal early exercise boundary. This has led Andersen and Broadie to propose a procedure that provides an upper bound to the price.[32] This procedure can be used in conjunction with any algorithm that generates a lower bound and pinpoints the true value of an American-style option more precisely than the algorithm does by itself.

SUMMARY

A number of models have been developed to fit the volatility smiles that are observed in practice. The constant elasticity of variance model leads to a volatility smile similar to that observed for equity options. The jump–diffusion model leads to a volatility smile similar to that observed for currency options. Variance-gamma and stochastic volatility models are more flexible in that they can lead to either the type of volatility smile observed for equity options or the type of volatility smile observed for currency options. The implied volatility function model provides even more flexibility than this. It is designed to provide an exact fit to any pattern of European option prices observed in the market.

The natural technique to use for valuing path-dependent options is Monte Carlo simulation. This has the disadvantage that it is fairly slow and unable to handle American-style derivatives easily. Luckily, trees can be used to value many types of path-dependent derivatives. The approach is to choose representative values for the underlying path function at each node of the tree and calculate the value of the derivative for each of these values as we roll back through the tree.

The binomial tree methodology can be extended to value convertible bonds. Extra branches corresponding to a default by the company are added to the tree. The roll-back calculations then reflect the holder's option to convert and the issuer's option to call.

Trees can be used to value many types of barrier options, but the convergence of the option value to the correct value as the number of time steps is increased tends to be slow. One approach for improving convergence is to arrange the geometry of the tree so that nodes always lie on the barriers. Another is to use an interpolation scheme to adjust for the fact that the barrier being assumed by the tree is different from the true barrier. A third is to design the tree so that it provides a finer representation of movements in the underlying asset price near the barrier.

One way of valuing options dependent on the prices of two correlated assets is to apply a transformation to the asset price to create two new uncorrelated variables.

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[32] See L. B. G. Andersen and M. Broadie, "A Primal-Dual Simulation Algorithm for Pricing Multi-Dimensional American Options," *Management Science*, 50, 9 (2004), 1222–34.

These two variables are each modeled with trees and the trees are then combined to form a single three-dimensional tree. At each node of the tree, the inverse of the transformation gives the asset prices. A second approach is to arrange the positions of nodes on the three-dimensional tree to reflect the correlation. A third approach is to start with a tree that assumes no correlation between the variables and then adjust the probabilities on the tree to reflect the correlation.

Monte Carlo simulation is not naturally suited to valuing American-style options, but there are two ways it can be adapted to handle them. The first uses a least-squares analysis to relate the value of continuing (i.e, not exercising) to the values of relevant variables. The second involves parameterizing the early exercise boundary and determining it iteratively by working back from the end of the life of the option to the beginning.

FURTHER READING

Andersen, L. B. G., "A Simple Approach to the Pricing of Bermudan Swaptions in the Multifactor LIBOR Market Model," *Journal of Computational Finance*, 3, 2 (Winter 2000): 1–32.

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